# Physical Priors and Real/Complex Representations in Reduced Catheter Prediction

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**Abstract**

**Purpose:** We separate physical prior and real/complex representation effects in reduced catheter prediction, and examine integration trade-offs and local invariant tori.

**Methods:** A factorial comparison uses real/complex Hamiltonian networks with and without a fixed quadratic prior. The 96 trajectories are split 64/16/16 for training, validation and testing; six groups, sixteen test trajectories and five seeds per model support five-effect analyses. An independent explicit conservative polynomial $H_{\mathrm{dec}}$ supports comparisons of central fourth order, midpoint, RK4 and two-stage fourth-order Gauss–Legendre, plus normal-form analysis.

**Results:** The prior improves prediction under both parameterizations, retaining its direction across prescribed weighting, aggregation, leave-one and group/seed resampling procedures. The sealed analysis did not resolve additional complex benefit. New sensitivity analysis fixed after the experiment gives a negative gray-box real-minus-complex error contrast, favouring real parameterization; no-prior representation and interaction remain unresolved. Central fourth order shows smaller energy changes than RK4 but requires approximately 2.59–4.02 times its integration time at selected state-accuracy targets. Its cost ordering against Gauss changes with the target. Exact quartic normal form and nondegeneracy establish qualitative local KAM tori for the polynomial at sufficiently small amplitudes.

**Conclusions:** The contribution is prior/representation separation within the fixed prediction design. Intervals depend on finite-design sampling assumptions and establish neither population coverage nor universal real/complex superiority. The existing central method offers an energy–cost trade-off, not a new algorithm or uniform performance advantage. Independent polynomial results do not certify learned or material systems, a numerical KAM amplitude, or specific invariant tori. These distinctions support restricted modelling comparisons and numerical choices.

**Keywords**

Hamiltonian neural networks, physical priors, complex representations, catheter modeling, symplectic integration, invariant tori

## 1 Research problem and contributions

### *1.1 Reduced prediction and the attribution problem*

Flexible-catheter prediction requires a low-dimensional state that retains the mechanics relevant to the intended motion. Spatial rod theory represents a deforming body by a reference curve and cross-sectional directions (Antman, 2005, Chapter 8); finite-strain formulations provide a basis for large-displacement beam dynamics (Simo, 1985). Magnetic interactions supply the actuation loads (Abbott et al., 2020). These mechanical descriptions motivate reduced states, but a predictor must still specify which coordinates, energy terms and external inputs it retains.

This paper studies finite-time state prediction in an existing reduced-catheter dataset. The primary data comprise 96 COMSOL reference trajectories, partitioned by complete trajectories into 64 training, 16 validation and 16 test trajectories. The tests belong to six orbit groups, and each model retains five training seeds. Learning uses a fixed four-mode basis, a constant mass matrix and canonical momentum $p = \boldsymbol{M}_4\dot{q}$. Different phases within one group and successive samples along a trajectory

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are correlated; the number of saved states is therefore not the number of independent cases. Our subject is comparative evidence within this fixed design, rather than new material identification, clinical control evaluation or safety certification across operating conditions.

A predictor contains at least three choices that can be varied separately: how much known energy structure is fixed, how its unknown part is parameterized, and how the resulting vector field is integrated. Changing all three together makes improvements difficult to attribute. We therefore compare the physical prior and real/complex representation factorially, while studying integrators on a separately defined conservative polynomial. This separation makes each question answerable within its own scope and requires an explicit account of the links not yet established between the two objects.

For the present comparison, a fixed quadratic energy $H_2$ supplies the prescribed prior, and a learned residual supplies its correction. This is a model choice: a lower prediction error with $H_2$ would show that the prior helps this task, not that it exactly identifies all terms of the material energy. The modal balancing of position and momentum determines the generation rule, whereas training-set scales determine the weighting of prediction errors. Section 2.1 fixes these conventions before either factor is varied.

### 1.2 Separating the prior from the representation

Hamiltonian neural networks generate a field from the symplectic gradient of a real scalar (Greydanus et al., 2019). The construction constrains the field while leaving its approximation to the observed dynamics to be learned. Port-Hamiltonian extensions add explicit dissipation and input channels (Zhong et al., 2020); the four Hamiltonian networks compared here close those channels. Their common generation rule therefore makes prior and representation comparisons interpretable without identifying the reference material system as conservative.

The physical prior has a specific meaning here: a fixed quadratic energy $H_2$ is added to a learnable residual energy. A model without the prior must fit the corresponding structure from data, whereas its gray-box counterpart learns a correction under the same generation principle. The question is not whether physical knowledge is useful in the abstract. It is how adding $H_2$ changes prediction errors for each parameterization under the same data split, metric and training endpoint. Even a reduction in finite test error does not establish that the prior equals the complete material energy.

Complex networks offer a different internal parameterization: their realified linear layers have coupled block weights (Trabelsi et al., 2018). Our implementation uses $Wz + b$, separate real/imaginary $\tanh$ and real-part readout. The learned object is consequently a real energy, although its internal features are complex. A scalar $H + iG$ with a specified generation rule is a separate representation question, treated in Section 2.5; the returned imaginary output in these experiments is not an identified dissipative potential.

Approximation theory does not replace a fixed-budget comparison. The ability of a growing smooth network family to approximate functions and derivatives (Hornik, 1991) differs from the error achieved with a prescribed width, optimizer and dataset. The real networks have hidden width 64 and the complex networks width 45. Approximate parameter matching does not equalize architecture constraints, optimization paths or runtime. A representation effect in this paper therefore compares these concrete implementations; it does not attribute every difference to the abstract number field.

We separate the factors using real/no-prior R, real/gray-box GR, complex/no-prior C and complex/gray-box GC in a $2 \times 2$ design. Prior effects are $\mathrm{R} - \mathrm{GR}$ and $\mathrm{C} - \mathrm{GC}$, representation effects are $\mathrm{R} - \mathrm{C}$ and $\mathrm{GR} - \mathrm{GC}$, and the interaction is $(\mathrm{GR} - \mathrm{GC}) - (\mathrm{R} - \mathrm{C})$. A residual MLP is a separate descriptive baseline and does not enter these five effects. The original trajectory-level analysis and the post-experiment group and seed sensitivity analysis use different sampling units and are reported separately; the latter protocol was fixed before its calculations. The questions concern directional stability, resolved differences and remaining uncertainty.

### 1.3 Independent questions about integration and conservative structure

A learned field must be advanced in time, so approximation quality and numerical accuracy answer different questions. Generating-function and symplectic Runge–Kutta methods already provide systematic ways to preserve the two-form (Feng et al., 1989; Hairer et al., 2006). We use the existing central fourth-order construction of Feng and colleagues, deriving the coefficients and solver interface needed to assess its cost. An exact implicit root can be symplectic without conserving the original energy exactly, and a finite solver tolerance does not itself prove exact symplecticity of a program. A state-error target must

therefore be accompanied by energy and computational-cost criteria.

We compare central fourth order, midpoint, RK4 and two-stage Gauss on one explicit conservative polynomial $H_{\mathrm{dec}}$. This is an independent numerical study: the prediction experiment retains RK4 and the learned $H_{\boldsymbol{\theta}}$. The common polynomial allows the integrators to be compared under identical initial conditions, state targets and cost accounting. It also permits a separate structural question: does its local normal form imply invariant tori of the full polynomial flow?

For that question, the tori of an integrable quartic truncation are only a starting point. The full analytic remainder must be controlled, and the action-frequency map must be nondegenerate before a persistence theorem can be used (Bounemoura, 2024; Pöschel, 2001). Section 4 gives this adaptation and obtains a qualitative small-amplitude family for $H_{\mathrm{dec}}$. The integrator comparison and this proof share the same polynomial, but address finite-time numerical choice and exact-flow structure, respectively. Neither validates the prediction effects or supplies the missing relation between the learned and material energies.

### 1.4 Findings and the argument of the paper

The primary questions are whether the fixed prior improves prediction and which real/complex differences can be resolved within each prior level. In the fixed design, both prior effects retain their improvement direction under the prescribed weighting, aggregation, leave-one and group/seed sensitivity procedures. The sealed analysis did not resolve additional complex benefit. The post-experiment sensitivity analysis gives a negative gray-box real-minus-complex error contrast, favouring the real model; no-prior representation and interaction remain unresolved. The contribution is this separation of effects and their sensitivity within the specified task, not a universal ordering of real and complex networks.

The two independent polynomial results have narrower roles. The numerical comparison establishes a state-accuracy, energy and cost trade-off for an existing algorithm, with no consistent advantage over RK4 and Gauss. The quartic normal form and its remainder support a model-specific application of classical local KAM theory, without a certified amplitude or learned/material torus. Their value is to distinguish practical numerical selection from structural persistence, not to add two algorithmic innovations.

The argument proceeds in three steps. Section 2 fixes the objects and derives the methods and actual energy-generation rule. Section 3 establishes the necessary numerical and approximation properties, then presents five-model descriptions, the four-model five effects and the independent integrator comparison. Section 4 uses the polynomial's exact normal form to address persistence. Sections 5 and 6 turn these results into a sequence of modelling choices and answer the research questions. The supplements retain the derivations and complete result files needed to reproduce the stated objects and comparisons.

## 2 Central symplectic schemes and complex Hamiltonian parameterization

Physical priors, energy parameterization, and time stepping determine, respectively, the prescribed model structure, the scalar to be learned, and the prediction obtained from its vector field. To evaluate these choices separately, we first specify the actual learning model and the independent conservative comparison, derive a central generating scheme from Hamilton's equations, and then establish the complex-coordinate rule used by the network. The fourth-order scheme is an existing generating-function method Feng et al. (1989); Guo et al. (2001). We give its complete construction in our notation rather than claim a new integration algorithm.

### 2.1 Two study objects and canonical notation

All states are column vectors. Bold symbols denote vectors and matrices; indexed components remain scalar. The subscripts on $H_{\boldsymbol{q}}$ and $H_{\boldsymbol{p}}$ denote column gradients. For $\boldsymbol{u} = (\boldsymbol{q}, \boldsymbol{p}) \in \mathbb{R}^{2n}$, define

$$\begin{gathered} \boldsymbol{J} = \begin{pmatrix} 0 & \boldsymbol{I}_n \\ -\boldsymbol{I}_n & 0 \end{pmatrix}, \quad \Omega = \sum_{j=1}^{n} \mathrm{d}q_j \wedge \mathrm{d}p_j, \\ \dot{\boldsymbol{u}} = \boldsymbol{J}\nabla H(\boldsymbol{u}) + \begin{pmatrix} 0 \\ \boldsymbol{r}(\boldsymbol{q}, \boldsymbol{p}, t) \end{pmatrix}. \end{gathered} \tag{1}$$

The momentum port $\boldsymbol{r}$ contains only forces not already included in $H$; a force generated by the potential is not added a second time. Both objects below have four degrees of freedom, but they are not assumed to share a known physical Hamiltonian.

The learning-coordinate contract uses a generalized modal matrix $\boldsymbol{U}$, satisfying in ideal exact arithmetic

$$\begin{gathered} \boldsymbol{U}^{\mathsf{T}}\boldsymbol{M}_4\boldsymbol{U} = \boldsymbol{I}, \quad \boldsymbol{U}^{\mathsf{T}}\boldsymbol{K}_4\boldsymbol{U} = \boldsymbol{\Lambda}^2, \\ \boldsymbol{S}_q = \boldsymbol{U}\boldsymbol{\Lambda}^{-1/2}, \quad \boldsymbol{S}_p = \boldsymbol{M}_4\boldsymbol{U}\boldsymbol{\Lambda}^{1/2}, \\ \boldsymbol{S} = \mathrm{diag}(\boldsymbol{S}_q, \boldsymbol{S}_p). \end{gathered} \tag{2}$$

where $\boldsymbol{\Lambda}$ is the positive diagonal modal-frequency matrix. Since $\boldsymbol{S}_q^{\mathsf{T}}\boldsymbol{S}_p = \boldsymbol{I}$, we have $\boldsymbol{S}^{\mathsf{T}}\boldsymbol{J}\boldsymbol{S} = \boldsymbol{J}$. The implemented prior in these coordinates is

$$H_2(\boldsymbol{Q}, \boldsymbol{P}) = \tfrac{1}{2}\boldsymbol{P}^{\mathsf{T}}\boldsymbol{M}_{\mathrm{norm}}^{-1}\boldsymbol{P} + \tfrac{1}{2}\boldsymbol{Q}^{\mathsf{T}}\boldsymbol{K}_{\mathrm{norm}}\boldsymbol{Q}. \tag{3}$$

**Table 1.** The two objects used in this section. Closing prediction ports does not imply that the material reference has no forcing or dissipation.

| | Actual learning experiment | Independent conservative comparison |
|---|---|---|
| State and momentum | $\boldsymbol{x} = (\boldsymbol{q}, \boldsymbol{p})$, $\boldsymbol{p} = \boldsymbol{M}_4\dot{\boldsymbol{q}}$; input $\boldsymbol{y} = (\boldsymbol{Q}, \boldsymbol{P}) = \boldsymbol{S}^{-1}\boldsymbol{x}$, with no momentum shift | $\boldsymbol{W} = (\boldsymbol{X}, \boldsymbol{Y})$; $\boldsymbol{Y}$ is the modal canonical momentum paired with $\boldsymbol{X}$, not the learning input $\boldsymbol{P}$ |
| Hamiltonian | Fixed quadratic prior $H_2$ and learned real scalar $H_{\boldsymbol{\theta}}$; the latter is not assumed to be the physical energy | Exact polynomial $H_{\mathrm{dec}}$ defined by stored decimal modal coefficients |
| Scales | Modal balancing $\boldsymbol{S}$; time in seconds, energy output factor 1, no division by a reference energy | A separate modal chart and the original time in seconds; no learning transform $\boldsymbol{S}$ or extra time scaling |
| Ports and stepping | Explicit ports of the four HNNs are closed; RK4 advances $\boldsymbol{J}\nabla_{\boldsymbol{y}} H_{\boldsymbol{\theta}}$ | $\boldsymbol{r} = 0$; central fourth order, midpoint, RK4, and two-stage fourth-order Gauss use the same $H_{\mathrm{dec}}$ |

Ideal balancing gives $\boldsymbol{M}_{\mathrm{norm}} = \boldsymbol{\Lambda}^{-1}$ and $\boldsymbol{K}_{\mathrm{norm}} = \boldsymbol{\Lambda}$. The implementation uses stored matrices numerically corresponding to this relation, not unit mass. For a real scalar $H_y$, if $H_x(\boldsymbol{x}) = H_y(\boldsymbol{S}^{-1}\boldsymbol{x})$, then

$$\nabla_{\boldsymbol{x}} H_x = \boldsymbol{S}^{-\mathsf{T}}\nabla_{\boldsymbol{y}} H_y, \qquad \dot{\boldsymbol{x}} = \boldsymbol{S}\boldsymbol{J}\nabla_{\boldsymbol{y}} H_y = \boldsymbol{J}\nabla_{\boldsymbol{x}} H_x. \tag{4}$$

Actual automatic differentiation acts on $\boldsymbol{y}$, not on the state $\boldsymbol{x}$ before preprocessing. The state scale in the loss weights prediction errors and does not introduce another canonical transformation. Modal balancing does not nondimensionalize time or energy; a complete unit identification between the material system and the energy output is not assumed here.

The independent comparison uses

$$\begin{aligned} H_{\mathrm{dec}}(\boldsymbol{W}) = {} & \tfrac{1}{2}\sum_{j=1}^{4}\omega_j(X_j^2 + Y_j^2) \\ & + \sum_{a\le b\le c} c^{(3)}_{abc} W_a W_b W_c \\ & + \sum_{a\le b\le c\le d} c^{(4)}_{abcd} W_a W_b W_c W_d. \end{aligned} \tag{5}$$

The cubic and quartic monomial coefficients are prescribed by the stored modal polynomial input, with decimal strings defining exact real numbers. The index convention and complete coefficient list accompany the supplementary material; see Section A.1. They are not reconstructed backwards from a target normal form. Defining this polynomial does not establish its identity with the complete material Hamiltonian or set material truncation errors to zero. The general method derived below can act on this specified object, whereas the learning experiment retains the generation and integration rules in Table 1.

## 2.2 From Hamilton's equations to a central generator

Take an autonomous real function $H : U \subset \mathbb{R}^{2n} \to \mathbb{R}$, let $\boldsymbol{f} = \boldsymbol{J}\nabla H$, and consider small steps and the local flow on a compact subdomain of $U$. For the finite fourth-order construction, assume $H \in C^8$ with the required bounded derivatives; a convergent infinite expansion requires analyticity separately. With the ports closed, Equation (1) becomes

$$\dot{\boldsymbol{q}} = H_{\boldsymbol{p}}, \qquad \dot{\boldsymbol{p}} = -H_{\boldsymbol{q}}. \tag{6}$$

For example, if $H(\boldsymbol{q}, \boldsymbol{p}) = \frac{1}{2}\boldsymbol{p}^{\mathsf{T}}\boldsymbol{N}(\boldsymbol{q})\boldsymbol{p} + V(\boldsymbol{q})$ with $\boldsymbol{N} = \boldsymbol{M}^{-1}$, then

$$\dot{\boldsymbol{q}} = \boldsymbol{N}(\boldsymbol{q})\boldsymbol{p}, \qquad \dot{p}_j = -\tfrac{1}{2}\boldsymbol{p}^{\mathsf{T}}\boldsymbol{N}_{,q_j}(\boldsymbol{q})\boldsymbol{p} - V_{,q_j}(\boldsymbol{q}). \tag{7}$$

Configuration-dependent mass terms and nonlinear potentials require their true gradients. In general, $\boldsymbol{K}(\boldsymbol{q})\boldsymbol{q}$ cannot replace $\nabla V$.

Let $\boldsymbol{u}^+ = \Phi_h(\boldsymbol{u}^-)$ be endpoints of the exact flow and define

$$\boldsymbol{m} = \tfrac{1}{2}(\boldsymbol{u}^+ + \boldsymbol{u}^-), \qquad \boldsymbol{\delta} = \boldsymbol{u}^+ - \boldsymbol{u}^-. \tag{8}$$

At $h = 0$, the derivative of the map from the old endpoint to $\boldsymbol{m}$ is the identity. Locally, the old endpoint can therefore be recovered from $\boldsymbol{m}$, and the increment can be written as $\boldsymbol{\delta} = \boldsymbol{F}(\boldsymbol{m}, h)$. With $\boldsymbol{A} = \mathrm{D}_{\boldsymbol{m}}\boldsymbol{F}$, we have $\mathrm{D}_{\boldsymbol{m}}\boldsymbol{u}^{\pm} = \boldsymbol{I} \pm \boldsymbol{A}/2$. Preservation of $\Omega$ by the exact flow gives, in center coordinates,

$$\begin{aligned} & (\boldsymbol{I} + \boldsymbol{A}/2)^{\mathsf{T}}\boldsymbol{J}(\boldsymbol{I} + \boldsymbol{A}/2) \\ & \quad - (\boldsymbol{I} - \boldsymbol{A}/2)^{\mathsf{T}}\boldsymbol{J}(\boldsymbol{I} - \boldsymbol{A}/2) = \boldsymbol{A}^{\mathsf{T}}\boldsymbol{J} + \boldsymbol{J}\boldsymbol{A} = 0. \end{aligned} \tag{9}$$

Thus $\mathrm{D}_{\boldsymbol{m}}(-\boldsymbol{J}\boldsymbol{F}) = -\boldsymbol{J}\boldsymbol{A}$ is symmetric. On a local star-shaped domain, the Poincaré lemma supplies a real scalar $\mathcal{S}_h$ such that

$$\nabla_{\boldsymbol{m}}\mathcal{S}_h = -\boldsymbol{J}\boldsymbol{F}, \qquad \boldsymbol{F} = \boldsymbol{J}\nabla_{\boldsymbol{m}}\mathcal{S}_h, \qquad \mathcal{S}_0 = 0. \tag{10}$$

The scalar $\mathcal{S}_h$ generates a time-$h$ map; it is neither the physical energy nor the complex network output.

To determine its coefficients, differentiation must hold $\boldsymbol{m}$ fixed. Along an exact trajectory with fixed old endpoint,

$\mathrm{d}\boldsymbol{\delta}/\mathrm{d}h = \boldsymbol{f}(\boldsymbol{u}^+)$ and $\mathrm{d}\boldsymbol{m}/\mathrm{d}h = \boldsymbol{f}(\boldsymbol{u}^+)/2$, so

$$\begin{aligned} \partial_h \boldsymbol{F} + \tfrac{1}{2}(\mathrm{D}_{\boldsymbol{m}}\boldsymbol{F})\boldsymbol{f}(\boldsymbol{m}+\boldsymbol{F}/2) &= \boldsymbol{f}(\boldsymbol{m}+\boldsymbol{F}/2), \\ \partial_h \boldsymbol{F} &= (\boldsymbol{I} - \tfrac{1}{2}\mathrm{D}_{\boldsymbol{m}}\boldsymbol{F})\boldsymbol{f}(\boldsymbol{m}+\boldsymbol{F}/2). \end{aligned} \tag{11}$$

Set $\boldsymbol{B} = \nabla^2_{\boldsymbol{m}}\mathcal{S}_h$. The chain rule yields

$$\begin{aligned} &\nabla_{\boldsymbol{m}} H(\boldsymbol{m} + \tfrac{1}{2}\boldsymbol{J}\nabla\mathcal{S}_h) \\ &\quad = (\boldsymbol{I} - \tfrac{1}{2}\boldsymbol{B}\boldsymbol{J})\nabla H(\boldsymbol{m} + \tfrac{1}{2}\boldsymbol{J}\nabla\mathcal{S}_h), \\ &\boldsymbol{J}\nabla_{\boldsymbol{m}} H(\boldsymbol{m} + \tfrac{1}{2}\boldsymbol{J}\nabla\mathcal{S}_h) \\ &\quad = (\boldsymbol{I} - \tfrac{1}{2}\boldsymbol{J}\boldsymbol{B})\boldsymbol{f}(\boldsymbol{m} + \tfrac{1}{2}\boldsymbol{J}\nabla\mathcal{S}_h). \end{aligned} \tag{12}$$

Substituting into Equation (11) and choosing the additive gauge depending only on $h$ gives the central Hamilton–Jacobi equation

$$\partial_h \mathcal{S}_h(\boldsymbol{m}) = H(\boldsymbol{m} + \tfrac{1}{2}\boldsymbol{J}\nabla\mathcal{S}_h(\boldsymbol{m})), \qquad \mathcal{S}_0 = 0. \tag{13}$$

This construction belongs to the established theory of canonical difference schemes Feng et al. (1989); Guo et al. (2001).

Keeping only $S_h^{[1]} = hH$ gives the midpoint scheme

$$\begin{aligned} \boldsymbol{u}^+ - \boldsymbol{u}^- &= h\boldsymbol{J}\nabla H(\boldsymbol{m}), \\ \boldsymbol{q}^+ - \boldsymbol{q}^- = hH_{\boldsymbol{p}}(\boldsymbol{m}), &\quad \boldsymbol{p}^+ - \boldsymbol{p}^- = -hH_{\boldsymbol{q}}(\boldsymbol{m}). \end{aligned} \tag{14}$$

It uses the full gradient in Equation (7) and does not require commuting mass and stiffness matrices. For an explicitly time-dependent $H$, every term is evaluated at $t_{\boldsymbol{m}} = t_n + h/2$. The autonomous fourth-order formula below cannot be extended merely by freezing time; the comparison here uses autonomous Hamiltonians.

## 2.3 Fourth-order coefficients and an executable update

Use ordinary power coefficients $\mathcal{S}_h = \sum_{j\ge 1} h^j S_j$, so that each $S_j$ already includes the Taylor factorial. Coefficient matching in Equation (13), with $[h^k]$ denoting coefficient extraction, gives

$$\begin{aligned} S_1 &= H, \\ S_{k+1} &= \frac{1}{k+1}[h^k]\, H\left(\boldsymbol{m} + \frac{1}{2}\boldsymbol{J}\sum_{j=1}^{k} h^j \nabla S_j(\boldsymbol{m})\right). \end{aligned} \tag{15}$$

This also defines finite Taylor coefficients without requiring convergence of an infinite series for a smooth Hamiltonian.

The first two comparisons are

$$2S_2 = \tfrac{1}{2}\nabla H^{\mathsf{T}}\boldsymbol{J}\nabla H = 0, \tag{16}$$

$$3S_3 = \tfrac{1}{2}\nabla H^{\mathsf{T}}\boldsymbol{J}\nabla S_2 + \tfrac{1}{8}H''[\boldsymbol{f},\boldsymbol{f}] = \tfrac{1}{8}H''[\boldsymbol{f},\boldsymbol{f}]. \tag{17}$$

The first nontrivial correction is therefore $S_3 = E/24$, where

$$E = \boldsymbol{f}^{\mathsf{T}}H''\boldsymbol{f}, \qquad \boldsymbol{f} = \boldsymbol{J}\nabla H. \tag{18}$$

To show that the next even coefficient also vanishes, differentiate $E$ along $\boldsymbol{f}$. Since $\boldsymbol{f}' = \boldsymbol{J}H''$,

$$\boldsymbol{f}\cdot\nabla E = H'''[\boldsymbol{f},\boldsymbol{f},\boldsymbol{f}] + 2(\boldsymbol{f}'\boldsymbol{f})^{\mathsf{T}}H''\boldsymbol{f} = H'''[\boldsymbol{f},\boldsymbol{f},\boldsymbol{f}], \tag{19}$$

The final quadratic term vanishes by skew-symmetry of $\boldsymbol{J}$. Matching the cubic power then gives

$$\begin{aligned} 4S_4 &= \tfrac{1}{2}\nabla H^{\mathsf{T}}\boldsymbol{J}\nabla S_3 + \tfrac{1}{48}H'''[\boldsymbol{f},\boldsymbol{f},\boldsymbol{f}] \\ &= -\tfrac{1}{48}\boldsymbol{f}\cdot\nabla E + \tfrac{1}{48}H'''[\boldsymbol{f},\boldsymbol{f},\boldsymbol{f}] = 0. \end{aligned} \tag{20}$$

The inverse relation $\Phi_{-h} = \Phi_h^{-1}$ for an autonomous flow gives the same parity: swapping endpoints leaves the center fixed and reverses the increment, permitting the gauge $\mathcal{S}_{-h} = -\mathcal{S}_h$.

The truncated generator and update are

$$\begin{aligned} S_h^{[3]}(\boldsymbol{m}) &= hH(\boldsymbol{m}) + \frac{h^3}{24}E(\boldsymbol{m}), \\ \boldsymbol{u}^+ - \boldsymbol{u}^- &= \boldsymbol{J}\nabla S_h^{[3]}(\boldsymbol{m}). \end{aligned} \tag{21}$$

To express the scalar formula through the vector field, differentiate each term:

$$\begin{aligned} \nabla E &= H'''[\,\cdot\,,\boldsymbol{f},\boldsymbol{f}] + 2\boldsymbol{f}'^{\mathsf{T}}H''\boldsymbol{f}, \\ \boldsymbol{J}\nabla E &= \boldsymbol{f}''[\boldsymbol{f},\boldsymbol{f}] - 2\boldsymbol{f}'^2\boldsymbol{f}, \\ \boldsymbol{g}_h &= \boldsymbol{J}\nabla S_h^{[3]} = h\boldsymbol{f} + \frac{h^3}{24}\big(\boldsymbol{f}''[\boldsymbol{f},\boldsymbol{f}] - 2\boldsymbol{f}'^2\boldsymbol{f}\big). \end{aligned} \tag{22}$$

Indeed, $\boldsymbol{J}H'''[\,\cdot\,,\boldsymbol{f},\boldsymbol{f}] = \boldsymbol{f}''[\boldsymbol{f},\boldsymbol{f}]$. Also, $\boldsymbol{f}' = \boldsymbol{J}H''$ and $(H'')^{\mathsf{T}} = H''$ give $\boldsymbol{J}\boldsymbol{f}'^{\mathsf{T}}H''\boldsymbol{f} = -\boldsymbol{J}H''\boldsymbol{J}H''\boldsymbol{f} = -\boldsymbol{f}'^2\boldsymbol{f}$, yielding the second line. Here $\boldsymbol{f}'^2\boldsymbol{f} = \boldsymbol{f}'(\boldsymbol{f}'\boldsymbol{f})$, and all terms are evaluated at the same center $\boldsymbol{m}$. The resulting method is

$$\boxed{\boldsymbol{u}^+ = \boldsymbol{u}^- + \boldsymbol{g}_h\left(\frac{\boldsymbol{u}^- + \boldsymbol{u}^+}{2}\right).} \tag{23}$$

After the corresponding coordinate and sign changes, this is the Type III central fourth-order formula of Feng et al. Feng et al. (1989), not an empirical correction to an arbitrary black-box vector field. Truncating at $h^3$ and obtaining order four are distinct statements: the exact central increment differs from $\boldsymbol{g}_h$ first at $h^5$, and stability of the local implicit

solve transfers that difference to the one-step error. Section 3 treats structure, local error, and fixed-finite-time global error separately.

For the solution interface, use the endpoint state $\boldsymbol{v} = \boldsymbol{u}^+$ as the unknown and define the residual in state units:

$$\begin{aligned} \mathcal{R}(\boldsymbol{v}; \boldsymbol{u}^-, h) &= \boldsymbol{v} - \boldsymbol{u}^- - \boldsymbol{g}_h((\boldsymbol{u}^- + \boldsymbol{v})/2), \\ \boldsymbol{B}(\boldsymbol{v}) &= \mathrm{D}_{\boldsymbol{v}}\mathcal{R} = \boldsymbol{I} - \tfrac{1}{2}\boldsymbol{J}\nabla^2 S_h^{[3]}((\boldsymbol{u}^- + \boldsymbol{v})/2), \\ \boldsymbol{B}(\boldsymbol{v}^{(k)})\Delta\boldsymbol{v}^{(k)} &= -\mathcal{R}(\boldsymbol{v}^{(k)}; \boldsymbol{u}^-, h), \\ \boldsymbol{v}^{(k+1)} &= \boldsymbol{v}^{(k)} + \alpha_k \Delta\boldsymbol{v}^{(k)}. \end{aligned} \tag{24}$$

The factor $0 < \alpha_k \leq 1$ allows backtracking when needed. The implementation solves the linear system rather than explicitly forming an inverse. A step starts from the near-identity predictor $\boldsymbol{v}^{(0)} = \boldsymbol{u}^- + h\boldsymbol{f}(\boldsymbol{u}^-)$, evaluates the true $H$ and its derivatives within the common domain, computes the Newton correction, and checks both the state residual and state correction. Nonconvergence or departure from the domain is a solver failure. Evaluation of $\boldsymbol{g}_h$ requires derivatives of $H$ through order three, while $\boldsymbol{B}$ requires order four; the full directional contractions are given in Section A.2.

On a convex neighborhood, if $\kappa = \frac{1}{2}\sup\|\mathrm{D}\boldsymbol{g}_h\| < 1$ and a radius $\rho$ satisfies $\|\boldsymbol{g}_h(\boldsymbol{u}^-)\| + \kappa\rho \leq \rho$, the fixed-point map takes that ball into itself and is a contraction, determining a unique local root. Newton convergence additionally requires an initial iterate within its convergence neighborhood. Local uniqueness does not exclude roots elsewhere, and a small residual does not prove global invertibility. The residual has state units and is not divided by $h$; its tolerance cannot be applied directly to stage slopes. The structure of the exact mathematical root differs from a finite-precision stopping rule: residual values control solution discrepancies, while structural defects also depend on the total residual derivative along the computed map; see Section A.3.

## 2.4 Complex coordinates and the actual real-energy network

For paired canonical coordinates $\boldsymbol{y} = (\boldsymbol{Q}, \boldsymbol{P})$, define

$$\boldsymbol{z} = \frac{\boldsymbol{Q} + \mathrm{i}\boldsymbol{P}}{\sqrt{2}}, \qquad \boldsymbol{Q} = \frac{\boldsymbol{z} + \bar{\boldsymbol{z}}}{\sqrt{2}}, \qquad \boldsymbol{P} = \frac{\boldsymbol{z} - \bar{\boldsymbol{z}}}{\mathrm{i}\sqrt{2}}. \tag{25}$$

The complex encoding acts on coordinate values with the specified scales; it does not assert physical nondimensionalization of the input. Treating $\boldsymbol{z}$ and $\bar{\boldsymbol{z}}$ as independent differentiation variables, the chain rule gives

$$\partial_{\boldsymbol{z}} = \frac{1}{\sqrt{2}}(\partial_{\boldsymbol{Q}} - \mathrm{i}\partial_{\boldsymbol{P}}), \qquad \partial_{\bar{\boldsymbol{z}}} = \frac{1}{\sqrt{2}}(\partial_{\boldsymbol{Q}} + \mathrm{i}\partial_{\boldsymbol{P}}). \tag{26}$$

For real $H$, $H_{\bar{\boldsymbol{z}}} = (H_{\boldsymbol{Q}} + \mathrm{i}H_{\boldsymbol{P}})/\sqrt{2}$. Hamilton's equations and the momentum port therefore combine as

$$\begin{aligned} \dot{\boldsymbol{z}} &= \frac{H_{\boldsymbol{P}} + \mathrm{i}(-H_{\boldsymbol{Q}} + \boldsymbol{r})}{\sqrt{2}} = -\mathrm{i}H_{\bar{\boldsymbol{z}}} + \frac{\mathrm{i}}{\sqrt{2}}\boldsymbol{r}, \\ \dot{\bar{\boldsymbol{z}}} &= \mathrm{i}H_{\boldsymbol{z}} - \frac{\mathrm{i}}{\sqrt{2}}\boldsymbol{r}. \end{aligned} \tag{27}$$

The second equation is the true conjugate of the first. The factors $\sqrt{2}$ and $-\mathrm{i}$ follow from Equation (25); they cannot be mixed with the convention $\boldsymbol{Q} + \mathrm{i}\boldsymbol{P}$.

The actual complex parameterization has three hidden layers and a readout layer:

$$\begin{aligned} \boldsymbol{a}_0 = \boldsymbol{z}, \quad \boldsymbol{a}_\ell &= \sigma(\boldsymbol{W}_\ell \boldsymbol{a}_{\ell-1} + \boldsymbol{b}_\ell)\ (\ell = 1, 2, 3), \\ F_{\boldsymbol{\theta}}(\boldsymbol{z}) &= \boldsymbol{W}_4 \boldsymbol{a}_3 + b_4, \\ \sigma(u + \mathrm{i}v) &= \tanh u + \mathrm{i}\tanh v. \end{aligned} \tag{28}$$

This is the W-only, `split_tanh` architecture: there is no term $\boldsymbol{V}_\ell \bar{\boldsymbol{a}}_{\ell-1}$, and the activation is not the holomorphic function $\tanh(\boldsymbol{z})$. For $\boldsymbol{W} = \boldsymbol{A} + \mathrm{i}\boldsymbol{B}$ and $\boldsymbol{b} = \boldsymbol{c} + \mathrm{i}\boldsymbol{d}$, each complex affine layer has the real form Trabelsi et al. (2018)

$$\begin{pmatrix} \mathrm{Re}(\boldsymbol{W}\boldsymbol{a} + \boldsymbol{b}) \\ \mathrm{Im}(\boldsymbol{W}\boldsymbol{a} + \boldsymbol{b}) \end{pmatrix} = \begin{pmatrix} \boldsymbol{A} & -\boldsymbol{B} \\ \boldsymbol{B} & \boldsymbol{A} \end{pmatrix} \begin{pmatrix} \mathrm{Re}\,\boldsymbol{a} \\ \mathrm{Im}\,\boldsymbol{a} \end{pmatrix} + \begin{pmatrix} \boldsymbol{c} \\ \boldsymbol{d} \end{pmatrix}. \tag{29}$$

The complex widths are $4 \to 45 \to 45 \to 45 \to 1$; the real network has widths $8 \to 64 \to 64 \to 64 \to 1$ with real $\tanh$ activations. Function classes and approximation conditions are discussed in Section 3; here we specify the functions that actually generate predictions.

Writing the real-network output as $h_{\boldsymbol{\theta}}(\boldsymbol{y})$, both parameterizations use

$$\begin{aligned} r_{\boldsymbol{\theta}}(\boldsymbol{y}) &= h_{\boldsymbol{\theta}}(\boldsymbol{y}) \quad \text{or} \quad \mathrm{Re}\,F_{\boldsymbol{\theta}}((\boldsymbol{Q} + \mathrm{i}\boldsymbol{P})/\sqrt{2}), \\ H_{\boldsymbol{\theta}}(\boldsymbol{y}) &= \beta H_2(\boldsymbol{y}) + e_s r_{\boldsymbol{\theta}}(\boldsymbol{y}), \qquad \beta \in \{0, 1\}, \quad e_s = 1, \\ \boldsymbol{f}_{\boldsymbol{\theta}}(\boldsymbol{y}) &= \boldsymbol{J}\nabla_{\boldsymbol{y}} H_{\boldsymbol{\theta}}(\boldsymbol{y}), \qquad \widehat{\boldsymbol{y}}_{k+1} = \mathrm{RK4}_h(\boldsymbol{f}_{\boldsymbol{\theta}})(\boldsymbol{y}_k). \end{aligned} \tag{30}$$

Here $\beta = 1$ denotes the gray-box prior and $\beta = 0$ the HNN without that prior; $H_2$ is added once. The actual energy output factor is $e_s = 1$, not the reference energy listed elsewhere in the data contract. Generating a field by an in-graph gradient of a real scalar follows the Hamiltonian neural-network construction Greydanus et al. (2019); it does not learn the central generator $S_h^{[3]}$. Training uses an RK4 one-step map error. Adding a constant $H_{\boldsymbol{\theta}} + c$ changes neither this vector field nor the map loss, so that loss does not identify the physical energy zero.

Real automatic differentiation with respect to $\boldsymbol{y}$ propagates the chain rule through complex encoding, realification, activation, and real readout. Differentiation with respect

to the original state must also use Equation (4). Internal imaginary channels can influence the real output and are updated through its loss gradient. However, the returned $\operatorname{Im} F_{\boldsymbol{\theta}}$ has no independent loss and does not enter the dynamics as a dissipative potential. Thus $F_{\boldsymbol{\theta}}$ is an internal complex parameterization and $H_{\boldsymbol{\theta}}$ the real scalar used for prediction; neither is the time-step central generator.

### 2.5 Compatibility and coordinate dependence of a single complex scalar

Representing ports through an imaginary part requires a specified generation rule, not merely writing two real functions as a complex number. Consider

$$\mathcal{H} = H + \mathrm{i}G, \qquad \dot{\boldsymbol{z}} = -\mathrm{i}\partial_{\bar{\boldsymbol{z}}}\mathcal{H}, \qquad \dot{\bar{\boldsymbol{z}}} = \mathrm{i}\partial_{\boldsymbol{z}}\overline{\mathcal{H}}, \tag{31}$$

with real $H, G$. The dependence of complex-Hamiltonian descriptions on the generation rule and complex structure also appears in dissipative-mechanics constructions Rajeev (2007). Expanding with Equation (26) gives

$$\begin{aligned} \partial_{\bar{\boldsymbol{z}}}(H + \mathrm{i}G) &= \frac{H_{\boldsymbol{Q}} - G_{\boldsymbol{P}} + \mathrm{i}(H_{\boldsymbol{P}} + G_{\boldsymbol{Q}})}{\sqrt{2}}, \\ \dot{\boldsymbol{Q}} &= H_{\boldsymbol{P}} + G_{\boldsymbol{Q}}, \quad \dot{\boldsymbol{P}} = -H_{\boldsymbol{Q}} + G_{\boldsymbol{P}}. \end{aligned} \tag{32}$$

This rule generates $\boldsymbol{J}\nabla H + \nabla G$; its Euclidean gradient component depends on the chosen coordinates.

**Proposition 2.1.** Single-scalar representation in fixed coordinates. *Fix the real energy $H$ on a local product domain with connected $\boldsymbol{Q}$ domain and star-shaped $\boldsymbol{P}$ domain, and let $\boldsymbol{r} \in C^1$. A real $G \in C^2$ makes Equation (31) equivalent to $\dot{\boldsymbol{Q}} = H_{\boldsymbol{P}}$, $\dot{\boldsymbol{P}} = -H_{\boldsymbol{Q}} + \boldsymbol{r}(\boldsymbol{Q}, \boldsymbol{P}, t)$ if and only if*

$$\mathrm{D}_{\boldsymbol{Q}}\boldsymbol{r} = 0, \qquad \mathrm{D}_{\boldsymbol{P}}\boldsymbol{r} = (\mathrm{D}_{\boldsymbol{P}}\boldsymbol{r})^{\mathsf{T}}. \tag{33}$$

*Time is a parameter here. On a general nonsimply connected domain, the corresponding closed-loop integrals must also vanish.*

**Proof.** Comparison with Equation (32) requires $G_{\boldsymbol{Q}} = 0$ and $G_{\boldsymbol{P}} = \boldsymbol{r}$, proving necessity. Conversely, let $\boldsymbol{P}_0$ be the star center and set

$$G(\boldsymbol{P}, t) = g_0(t) + \int_0^1 \boldsymbol{r}(\boldsymbol{P}_0 + s(\boldsymbol{P} - \boldsymbol{P}_0), t)^{\mathsf{T}}(\boldsymbol{P} - \boldsymbol{P}_0)\,\mathrm{d}s. \tag{34}$$

Differentiation with respect to $P_j$, using $\partial_{P_j} r_i = \partial_{P_i} r_j$, turns the integrand into $\mathrm{d}\{s r_j(\boldsymbol{P}_0 + s(\boldsymbol{P} - \boldsymbol{P}_0), t)\}/\mathrm{d}s$. Hence $G_{P_j} = r_j(\boldsymbol{P}, t)$ and $G_{\boldsymbol{Q}} = 0$, proving sufficiency. Globally, the periods of the same one-form $\boldsymbol{r} \cdot \mathrm{d}\boldsymbol{P}$ must vanish. □

For example, $\boldsymbol{r} = -\boldsymbol{\Gamma}(t)\boldsymbol{P} + \boldsymbol{b}(t)$ with $\boldsymbol{\Gamma} = \boldsymbol{\Gamma}^{\mathsf{T}}$ admits $G = -\boldsymbol{P}^{\mathsf{T}}\boldsymbol{\Gamma}\boldsymbol{P}/2 + \boldsymbol{b}^{\mathsf{T}}\boldsymbol{P}$. General position-dependent damping does not satisfy these conditions. A force with a position potential should be included once in the true total $H$, or retained as an explicit port, rather than automatically assigned to $G$. Dissipation separately requires $H_{\boldsymbol{P}}^{\mathsf{T}}\boldsymbol{r} \le 0$; it does not follow from the sign of the imaginary part or integrability alone.

Coordinate choice can change gradient compatibility without changing the value of the same physical energy. Consider a mechanical system already expressed in consistently scaled canonical variables, without momentum shift, with $H(\boldsymbol{q}, \boldsymbol{p}) = \frac{1}{2}\boldsymbol{p}^{\mathsf{T}}\boldsymbol{M}^{-1}\boldsymbol{p} + V(\boldsymbol{q})$, constant $\boldsymbol{M} \succ 0$, $\boldsymbol{C} = \boldsymbol{C}^{\mathsf{T}} \succeq 0$, and port $-\boldsymbol{C}\boldsymbol{M}^{-1}\boldsymbol{p}$. Put $\boldsymbol{L} = \boldsymbol{M}^{1/2}$; then

$$\begin{aligned} \boldsymbol{Q} = \boldsymbol{L}\boldsymbol{q}, \quad \boldsymbol{P} = \boldsymbol{L}^{-1}\boldsymbol{p}, \\ \boldsymbol{T} = \operatorname{diag}(\boldsymbol{L}, \boldsymbol{L}^{-1}), \quad \boldsymbol{T}^{\mathsf{T}}\boldsymbol{J}\boldsymbol{T} = \boldsymbol{J}, \\ \bar{H}(\boldsymbol{Q}, \boldsymbol{P}) = \tfrac{1}{2}\boldsymbol{P}^{\mathsf{T}}\boldsymbol{P} + V(\boldsymbol{L}^{-1}\boldsymbol{Q}), \\ \bar{\boldsymbol{C}} = \boldsymbol{L}^{-1}\boldsymbol{C}\boldsymbol{L}^{-1} = \bar{\boldsymbol{C}}^{\mathsf{T}} \succeq 0. \end{aligned} \tag{35}$$

The chain rule gives $\dot{\boldsymbol{Q}} = \boldsymbol{P}$ and $\dot{\boldsymbol{P}} = -\bar{H}_{\boldsymbol{Q}} - \bar{\boldsymbol{C}}\boldsymbol{P}$. Thus $G = -\boldsymbol{P}^{\mathsf{T}}\bar{\boldsymbol{C}}\boldsymbol{P}/2$ in the new complex coordinates $\boldsymbol{\zeta} = (\boldsymbol{Q} + \mathrm{i}\boldsymbol{P})/\sqrt{2}$ recovers a single-scalar representation without requiring $\boldsymbol{M}$ and $\boldsymbol{C}$ to commute. Its real part is just the original $H$ in new coordinates, with no auxiliary energy added. However, if the old coordinate is $\boldsymbol{\xi} = (\boldsymbol{q} + \mathrm{i}\boldsymbol{p})/\sqrt{2}$, then

$$\boldsymbol{\zeta} = \tfrac{1}{2}(\boldsymbol{L} + \boldsymbol{L}^{-1})\boldsymbol{\xi} + \tfrac{1}{2}(\boldsymbol{L} - \boldsymbol{L}^{-1})\bar{\boldsymbol{\xi}}. \tag{36}$$

This transformation generally mixes a complex coordinate and its conjugate. Choosing the standard Euclidean gradient in the new coordinates changes the gradient metric associated with the original complex structure. The obstruction from nonsymmetric $\boldsymbol{C}\boldsymbol{M}^{-1}$ in the old fixed coordinates remains valid. This mass normalization is also different from the actual modal balancing in Equation (2); it does not establish that actual damping belongs to this subclass.

Appendix A.4 gives fixed-coordinate counterexamples and the limits of extending the mass normalization. The actual network uses the real-energy rule in (30); the single-scalar proposition specifies a separate representation boundary. We now analyse the approximation and numerical properties needed to interpret the comparisons.

## 3 Numerical properties, approximation and comparative results

Section 2 separates two modelling uses: the prediction models generate vector fields from learned real energies, whereas the central generating scheme discretizes the explicitly defined conservative polynomial $H_{\mathrm{dec}}$. We first analyse the exact mathematical map, then the expressive scope of the implemented networks, and finally the roles of prior information and representation in prediction and the error–cost trade-offs of integration. The RK4 implementation used for learning is distinct from the central fourth-order method compared here.

### 3.1 Single-valued maps, exact symplecticity and error propagation

Using (21)–(23), write $S = S_h^{[3]}$, $\boldsymbol{g}_h = \boldsymbol{J}\nabla S$, and let $\boldsymbol{v} = \Psi_h(\boldsymbol{u})$ denote the near-identity root of (23), with $\boldsymbol{m} = (\boldsymbol{u} + \boldsymbol{v})/2$.

The following are properties of the central generating-function method (Feng et al., 1989; Guo et al., 2001); small observed energy changes are not their proof.

**Proposition 3.1.** *Let $H$ be $C^8$ on a domain containing the centres used below and their neighbourhood. Suppose that (23) has a uniformly selected single-valued near-identity branch and that $\boldsymbol{B} = \boldsymbol{I} - \mathrm{D}\boldsymbol{g}_h(\boldsymbol{m})/2$ is invertible. Its map preserves the full symplectic two-form. If $S$ is single-valued on this domain, the map is also exact symplectic with respect to $\alpha = \boldsymbol{p}\cdot\mathrm{d}\boldsymbol{q}$.*

**Proof.** Set $\boldsymbol{A} = \mathrm{D}\boldsymbol{g}_h(\boldsymbol{m}) = \boldsymbol{J}\nabla^2 S(\boldsymbol{m})$. Symmetry of the Hessian gives $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{J} + \boldsymbol{J}\boldsymbol{A} = 0$. Differentiating the implicit equation yields

$$\begin{aligned} \boldsymbol{B}\mathrm{D}\Psi_h = \boldsymbol{C}, \quad \boldsymbol{C} = \boldsymbol{I} + \boldsymbol{A}/2, \\ \mathrm{D}\Psi_h = \boldsymbol{B}^{-1}\boldsymbol{C}, \quad (\mathrm{D}\Psi_h)^{\mathsf{T}}\boldsymbol{J}\mathrm{D}\Psi_h = \boldsymbol{J}. \end{aligned} \tag{37}$$

The final equality follows directly from

$$\begin{aligned} \boldsymbol{B}^{\mathsf{T}}\boldsymbol{J}\boldsymbol{B} = \boldsymbol{C}^{\mathsf{T}}\boldsymbol{J}\boldsymbol{C} = \boldsymbol{J} + \tfrac{1}{4}\boldsymbol{A}^{\mathsf{T}}\boldsymbol{J}\boldsymbol{A}, \quad \boldsymbol{B}\boldsymbol{C} = \boldsymbol{C}\boldsymbol{B}, \\ (\mathrm{D}\Psi_h)^{\mathsf{T}}\boldsymbol{J}\mathrm{D}\Psi_h = \boldsymbol{B}^{-\mathsf{T}}\boldsymbol{C}^{\mathsf{T}}\boldsymbol{J}\boldsymbol{C}\boldsymbol{B}^{-1} = \boldsymbol{J}. \end{aligned} \tag{38}$$

Thus the entire two-form is checked, not merely the determinant. For a single-valued primitive, write $\boldsymbol{\delta} = \boldsymbol{v} - \boldsymbol{u} = (\boldsymbol{\delta_q}, \boldsymbol{\delta_p})$. Since $\boldsymbol{\delta_q} = S_{\boldsymbol{m_p}}$ and $\boldsymbol{\delta_p} = -S_{\boldsymbol{m_q}}$,

$$\begin{aligned} \boldsymbol{p}^+\mathrm{d}\boldsymbol{q}^+ - \boldsymbol{p}\,\mathrm{d}\boldsymbol{q} &= \boldsymbol{m_p}\cdot\mathrm{d}\boldsymbol{\delta_q} + \boldsymbol{\delta_p}\cdot\mathrm{d}\boldsymbol{m_q} \\ &= \mathrm{d}(\boldsymbol{m_p}\cdot\boldsymbol{\delta_q}) - \mathrm{d}S(\boldsymbol{m}), \\ \Psi_h^*\alpha - \alpha &= \mathrm{d}Q_h, \\ Q_h(\boldsymbol{u}) &= \boldsymbol{m_p}(\boldsymbol{u})\cdot\boldsymbol{\delta_q}(\boldsymbol{u}) - S(\boldsymbol{m}(\boldsymbol{u})). \end{aligned} \tag{39}$$

Here $\mathrm{d}\alpha = -\Omega$, consistently with $\Omega = \sum_i dq_i \wedge dp_i$. Single-valuedness of $S$ and of the branch makes $Q_h$ single-valued; local closedness alone does not establish this property. □

A useful sufficient condition is to place a closed ball of radius $a$ around each $\boldsymbol{u}$, keep all centres in a common domain, and require

$$\kappa = \tfrac{1}{2}\sup\|D\boldsymbol{g}_h\| < 1, \qquad \|\boldsymbol{g}_h(\boldsymbol{u})\| + \kappa a < a. \tag{40}$$

The right-hand side is then a contraction into that ball. Uniqueness in a uniform near-identity neighbourhood identifies the branches on overlaps. This constructs a local map and its derivative, but does not by itself make the entire annulus invariant. If $H$ is analytic and the bounds hold on a common complex domain, the selected branch is analytic.

**Proposition 3.2.** *Suppose that exact and numerical trajectories remain in a common compact neighbourhood for fixed $0 \le nh \le T$, and that (40) holds uniformly for sufficiently small $h > 0$. The central fourth-order map has local error $O(h^5)$ and global error $O(h^4)$ on this fixed time interval.*

**Proof.** The odd expansion of the exact central generator and the truncation in Section 2 give $\Delta_h(\boldsymbol{m}) = \boldsymbol{g}_h(\boldsymbol{m}) + O(h^5)$, with uniform bounds on the remainder and the derivatives needed on the common neighbourhood. Substitution of the exact endpoint into (23) therefore leaves an $O(h^5)$ residual. The contraction estimate converts this residual into an endpoint error:

$$\|\Psi_h(\boldsymbol{u}) - \Phi_h(\boldsymbol{u})\| \le \frac{C_0 h^5}{1-\kappa}. \tag{41}$$

Since $\mathrm{D}\boldsymbol{g}_h = O(h)$, (37) implies $\|D\Psi_h\| \le (1+\kappa)/(1-\kappa) \le 1 + Lh$. Thus $e_n = \|\boldsymbol{u}_n - \Phi_{nh}(\boldsymbol{u}_0)\|$ satisfies

$$e_{n+1} \le (1+Lh)e_n + Ch^5, \qquad e_n \le \frac{C}{L}(\mathrm{e}^{LT} - 1)h^4 \tag{42}$$

with the limiting interpretation at $L = 0$. This is a fixed-$T$ estimate, not a trajectory bound uniform as $T \to \infty$, and does not prove the existence of KAM tori. □

Newton iteration has a nonzero residual. For an idealized $C^1$ execution map $\widetilde{\Psi}_h$, define the state residual, without division by $h$, by

$$\boldsymbol{r}_h(\boldsymbol{u}) = \widetilde{\Psi}_h(\boldsymbol{u}) - \boldsymbol{u} - \boldsymbol{g}_h((\boldsymbol{u} + \widetilde{\Psi}_h(\boldsymbol{u}))/2). \tag{43}$$

If its uniform bound is $\tau_0$, the one-step solution error is at most $\tau_0/(1-\kappa)$, and the global estimate becomes $C_T(h^4 + \tau_0/h)$. The sufficient scaling for fourth order is $\tau_0 = O(h^5)$; a fixed tolerance eventually limits the observable order. The full derivative-defect identity is given in Appendix A.3. A small residual value alone does not control this defect: bounds on its derivative and on the inverse matrix are also needed. Floating-point rounding and stopping branches need not even define an everywhere $C^1$ map. The tolerance checks below assess finite numerical resolution, not exact symplecticity of a program.

## 3.2 Approximation by the implemented network class

The block constraint in (29) does not make the implemented family equal to all real networks of the same width, but also does not exclude approximation of real functions. The following explicit embedding connects the actual architecture to the growing real-network family in Hornik's Theorem 3 (Hornik, 1991). The construction is an adaptation for this readout, not a claim of a new approximation theorem.

**Proposition 3.3.** *Under the convention $\boldsymbol{z} = (\boldsymbol{Q} + \mathrm{i}\boldsymbol{P})/\sqrt{2}$, every real $\tanh$ network can be embedded in the real output of the stated complex network with the same hidden widths. For the three-hidden-layer family with growing width, real functions that are $C^m$ on a neighbourhood of a compact set can be approximated in the $C^m$ norm, for any fixed finite integer $m$.*

**Proof.** For a real first layer $\boldsymbol{A}_Q\boldsymbol{Q} + \boldsymbol{A}_P\boldsymbol{P} + \boldsymbol{c}$, choose

$$\begin{aligned} \boldsymbol{W}_1 &= \sqrt{2}(\boldsymbol{A}_Q - \mathrm{i}\boldsymbol{A}_P), \quad \operatorname{Re}\boldsymbol{b}_1 = \boldsymbol{c}; \\ \boldsymbol{W}_\ell &= \boldsymbol{A}_\ell \in \mathbb{R}^{d_\ell \times d_{\ell-1}} \quad (\ell \geq 2). \end{aligned} \tag{44}$$

Take subsequent biases and readout weights real. The first real channel reproduces the prescribed affine function. The split activation and subsequent real weights then reproduce every real channel inductively; the imaginary channels need not vanish. Derivative approximation for growing networks follows from Hornik's smooth-activation theorem (Hornik, 1991). Starting with a one-hidden-layer approximant, scaled $\tanh$ layers approximate the identity on bounded sets and embed it into the three-hidden-layer family. The extension and norm conditions are detailed in Appendix B.1. □

The real and complex hidden widths are 64 and 45. The same-width embedding therefore proves neither containment of these two fixed architectures nor attainment of a best approximation by training. The generation rule and scales remain those in (30) and (4).

If $\|H_{\boldsymbol{\theta}} - H_*\|_{C^{r+1}}$ is known on a common domain, linearity of $\boldsymbol{J}\nabla$ bounds the $C^r$ vector-field error. For a fixed step and common bounded stages, the RK4 map also depends continuously on that field. Training uses a one-step map loss, for which the reverse implication fails. For example, in one degree of freedom,

$$H_k(q,p) = k^{-2}\sin(kq), \qquad \dot{q} = 0, \quad \dot{p} = -k^{-1}\cos(kq). \tag{45}$$

RK4 produces the exact translation, differing from the identity by at most $h/k$, whereas $\sup|\partial_q^4 H_k| = k^2$. An additive constant in $H$ also leaves the vector field unchanged. A small map loss thus fixes neither the energy gauge nor a small high-order or analytic error norm. Availability of high-order automatic derivatives, agreement at finitely many samples, and a global error certificate are separate assertions.

## 3.3 Five effects of prior information and representation

The primary dataset contains 96 reference trajectories, with a fixed whole-trajectory/orbit-group split of 64 training, 16 validation and 16 test trajectories. Each has 801 state points over 0–0.2 s at $\Delta t = 2.5 \times 10^{-4}$ s. Every seed uses the epoch-300 model; seeds and checkpoints are not selected by test performance.

The prediction comparison uses the existing 400 metric rows from 16 trajectories and 5 training seeds per model. The residual MLP among the five models is an external descriptive baseline. The factorial comparison comprises real R, complex C, and their $H_2$-prior counterparts GR and GC. Taking the median over five seeds for each model–trajectory pair yields 80 aggregated rows before trajectory averaging. The primary metric is the stored `rollout_mean_error`, retaining its prescribed normalization without logarithmic transformation. With training-only fixed loss scales $d_j$ and autonomous rollout from the true initial condition, the per-trajectory metric is

$$\begin{aligned} \boldsymbol{D}_{\mathrm{train}} &= \operatorname{diag}(d_1, \ldots, d_8), \\ X_{msi} &= \frac{1}{800}\sum_{k=1}^{800}\frac{1}{8}\left\|\boldsymbol{D}_{\mathrm{train}}^{-1}(\widehat{\boldsymbol{y}}_{msi,k} - \boldsymbol{y}_{i,k})\right\|_2^2. \end{aligned} \tag{46}$$

**Table 2.** Five-model descriptive rollout MSE. Each trajectory first takes the median over five seeds; columns summarize those 16 values. These are original trajectory-weighted descriptions, not group-weighted sensitivity estimates or significance tests.

| Model | Mean | Median |
|---|---|---|
| R | $1.66533 \times 10^{3}$ | $1.37061 \times 10^{3}$ |
| GR | $1.26587 \times 10^{1}$ | $9.57033 \times 10^{0}$ |
| C | $1.66528 \times 10^{3}$ | $1.37065 \times 10^{3}$ |
| GC | $1.26640 \times 10^{1}$ | $9.60739 \times 10^{0}$ |
| Residual MLP | $2.20953 \times 10^{9}$ | $2.20953 \times 10^{9}$ |

Table 2 retains all five models under the original common aggregation. The large Residual MLP values are retained as observed; this baseline predicts a residual map and does not enter the four-model Hamiltonian factorial contrasts. The descriptive ranking supplies no new significance claim. The five effects below isolate prior and representation.

This additional loss scaling is distinct from the canonical transformation $\boldsymbol{S}$, and the mean squared error is not a time-uniform supremum norm. The five effect directions are

$$\begin{aligned}\big(&\mathrm{R}-\mathrm{GR},\ \mathrm{C}-\mathrm{GC},\ \mathrm{R}-\mathrm{C},\ \mathrm{GR}-\mathrm{GC},\\ &(\mathrm{GR}-\mathrm{GC})-(\mathrm{R}-\mathrm{C})\big).\end{aligned} \tag{47}$$

Positive values in the first two contrasts favour the prior; positive values in the next two favour complex representation. The last contrast is the interaction between representation and prior.

The sealed analysis resamples paired trajectories conditional on the existing five-seed aggregation. The new sensitivity analysis retains that table and considers six orbit groups of sizes 2, 2, 4, 2, 2, 4, retaining all phases within each group and varying seed and group handling. Table 3 juxtaposes the sealed intervals and the new primary analysis: equal group weighting, seed medians, and joint group–seed resampling. They target different sources of randomness; a narrower interval does not directly mean a more precise estimate of the same estimand.

Both prior effects retain the improvement direction under the prescribed two weighting rules, seed mean/median aggregation, all leave-one checks, and three resampling schemes. The new gray-box contrast $\mathrm{GR}-\mathrm{GC}$ is negative in its primary and other prescribed intervals, indicating lower error for GR in this design. The sealed analysis did not resolve this difference and remains reported. The no-prior contrast $\mathrm{R}-\mathrm{C}$ and the interaction still have intervals crossing zero. Whether separate simple contrasts resolve a difference is not a test of their interaction. These results establish neither universal real/complex superiority, equivalence, nor an absence of benefit.

The new protocol was fixed after the original experiment and is a sensitivity analysis, not a preregistered confirmation of that experiment. Resampling six groups and five seeds relies on working assumptions such as exchangeability. Population coverage of the nominal percentile and five-effect-adjusted intervals has not been proved. Appendix B.2 gives the aggregation, leave-one, random-stream and interpretation details. The $H_2$ `energy_error` and learned-$H_{\boldsymbol{\theta}}$ drift are distinct from this primary metric. The latter also depends on the energy baseline and floating-point cancellation. Finite records of zero drift do not replace a structural identity, and the sensitivity analysis does not substitute a new drift metric.

### 3.4 Integrator comparison on the same conservative model

We compare the central fourth-order method, implicit midpoint, classical RK4 and the two-stage fourth-order Gauss–Legendre method (Hairer et al., 2006), all applied to the same $H_{\mathrm{dec}}$ in (5), with its actual gradient and required higher derivatives. This comparison is independent of learning and measures state accuracy, energy changes and solution cost on a nonlinear conservative object.

Let $\boldsymbol{u}_1 = (1,1,1,1,0,0,0,0)/2$ and $\boldsymbol{u}_2 = (1,2,3,4,4,-3,2,-1)/\sqrt{60}$. The four initial conditions are

$$\begin{aligned}\boldsymbol{A}_1 &= (r/2)\boldsymbol{u}_1, \quad \boldsymbol{A}_2 = (r/2)\boldsymbol{u}_2,\\ \boldsymbol{B}_1 &= r\boldsymbol{u}_1, \quad \boldsymbol{B}_2 = r\boldsymbol{u}_2,\\ r &= 1.4810634592307795 \times 10^{-11}.\end{aligned} \tag{48}$$

The solver scale is $R = 1.0521287803311451 \times 10^{-10}$. It follows from a conservative local energy restriction on the polynomial and is not a certified KAM radius. The fixed grid is $h_j = h_0/2^j$, $j = 0,\dots,4$, with $h_0 = 0.0017810279035513454$ and $T = 2019h_0 = 3.5958953372701665$, approximately 20 slow linear periods. Initial absolute cubic/quadratic and quartic/quadratic energy ratios are $1.13 \times 10^{-5}$–$2.30 \times 10^{-5}$ and $3.97 \times 10^{-6}$–$2.01 \times 10^{-5}$, respectively. The tested regime is weakly nonlinear and does not establish large-amplitude performance.

Using a tightened independent DOP853 solution as reference, the primary common-grid error and the native-step secondary metric are

$$\begin{aligned}e_{\mathrm{state}} &= \max_t \frac{\|\boldsymbol{W}_{\mathrm{num}}(t) - \boldsymbol{W}_{\mathrm{ref}}(t)\|_2}{\|\boldsymbol{W}_0\|_2},\\ e_H &= \max_n \frac{|H_{\mathrm{dec}}(\boldsymbol{W}_n) - H_{\mathrm{dec}}(\boldsymbol{W}_0)|}{H_{2,\mathrm{dec}}(\boldsymbol{W}_0)}.\end{aligned} \tag{49}$$

**Table 3.** Five effects: sealed results and new primary sensitivity analysis. Original columns use equal trajectory weights and seed medians; new columns use equal group weights and seed medians. Sealed 95% intervals, new 95% intervals and nominal five-effect-adjusted intervals are retained separately. Values are rounded only for display.

| Effect | Original | New | Sealed 95% interval |
|---|---|---|---|
| $\mathrm{R}-\mathrm{GR}$ | 1652.68 | 1438.62 | [1043.01, 2314.78] |
| $\mathrm{C}-\mathrm{GC}$ | 1652.62 | 1438.58 | [1037.70, 2311.88] |
| $\mathrm{R}-\mathrm{C}$ | 0.05316 | 0.03448 | [-0.08880, 0.17862] |
| $\mathrm{GR}-\mathrm{GC}$ | -0.00532 | -0.00529 | [-0.14163, 0.12850] |
| $(\mathrm{GR}-\mathrm{GC})-(\mathrm{R}-\mathrm{C})$ | -0.05848 | -0.03977 | [-0.17323, 0.06401] |

| Effect | New 95% interval | New adjusted interval |
|---|---|---|
| $\mathrm{R}-\mathrm{GR}$ | [588.62, 2462.33] | [332.78, 2778.00] |
| $\mathrm{C}-\mathrm{GC}$ | [588.54, 2462.32] | [332.58, 2777.95] |
| $\mathrm{R}-\mathrm{C}$ | [-0.08579, 0.95059] | [-0.11404, 0.98582] |
| $\mathrm{GR}-\mathrm{GC}$ | [-0.01035, -0.00146] | [-0.01274, -0.00103] |
| $(\mathrm{GR}-\mathrm{GC})-(\mathrm{R}-\mathrm{C})$ | [-0.95552, 0.08142] | [-0.99095, 0.10964] |

The denominator $H_{2,\mathrm{dec}}$ is this object's quadratic part, not the fixed $H_2$ comparator in learning. The prescribed reference and Newton-tolerance checks satisfy their criteria, but do not provide rigorous arithmetic enclosures. Gauss uses 16 stage-state components as Newton unknowns, with residuals in $\boldsymbol{W}$ units. Both the residual and the final correction, divided by $R$, must be at most $10^{-13}$. A stage-slope tolerance is not interpreted as a state tolerance. Appendix B.3 records the implementation convention and reference settings.

Cost includes derivative assembly, Newton corrections, linear solves, line searches, native-step energy evaluation and counters, using float64 and one CPU thread in a common environment (Windows 11; AMD64 Family 25 Model 97 Stepping 2; Python 3.12.7, NumPy 1.26.4, SciPy 1.13.1 and Numba 0.60.0). Each configuration is warmed once and timed three times in a fixed randomized order. Shared preparation takes 57.0300 s, including approximately 56.9762 s of compilation. A one-off job adds this preparation to the plotted integration cost. Reference generation, file output and plotting are excluded from algorithm integration time. A local post-warm-up saving therefore does not directly imply the same saving for a one-off job.

For each target, the lowest measured cost among resolved grid points is selected without interpolating to untested step sizes. All 80 configurations completed. Of the 48 initial-condition–method–target combinations, 36 attained the target; all 12 midpoint combinations did not. This means that midpoint did not attain the targets on this fixed grid. Its measured finite costs remain recorded and are not replaced by infinity.

At $10^{-3}$, central fourth order has approximately 24% lower post-warm-up integration cost than Gauss on the fixed grid and implementation; the shared 57.0300 s preparation is additional. Both methods select $j = 2$ for all four initial conditions at this target. The selected central configurations have $e_H$ ranging from $1.94249 \times 10^{-10}$ to $2.78099 \times 10^{-10}$, versus $1.55388 \times 10^{-11}$ to $4.45665 \times 10^{-11}$ for Gauss, with a larger observed energy change for the central method in every paired initial condition. This compares the energy cost of configurations selected for the same state target; the difference is not an exact energy-error certificate. Gauss is faster at both $10^{-4}$ and $10^{-5}$. Relative to RK4, the central method costs approximately 2.59–4.02 times as much at the selected state-accuracy targets while showing smaller energy changes. The finding is a trade-off among state accuracy, energy behaviour and solution cost, not consistent performance superiority. At the finest step, both central fourth order and Gauss have $e_H$ of approximately $10^{-12}$–$10^{-11}$. Differences at this scale have not been reliably resolved, so these tiny energy errors are not ranked. Appendix B.3 retains every step-size energy and failure summary. Small energy changes on a finite window do not prove that the trajectories lie on KAM tori.

## 4 Normal form and local KAM tori of the explicit conservative model

The object of this section is $H_{\mathrm{dec}}$, defined by (5) and the decimal coefficients in Appendix A.1. It is an explicit real-analytic polynomial with four degrees of freedom, specified quadratic modal frequencies, and cubic and quartic terms. The learned energy $H_{\boldsymbol{\theta}}$, learning prior $H_2$, quartic normal form, and discrete maps are distinct objects. The result below establishes a positive-measure family of invariant tori of $H_{\mathrm{dec}}$ at sufficiently small amplitudes. It does not certify a numerical amplitude, a material system, or a particular learned model.

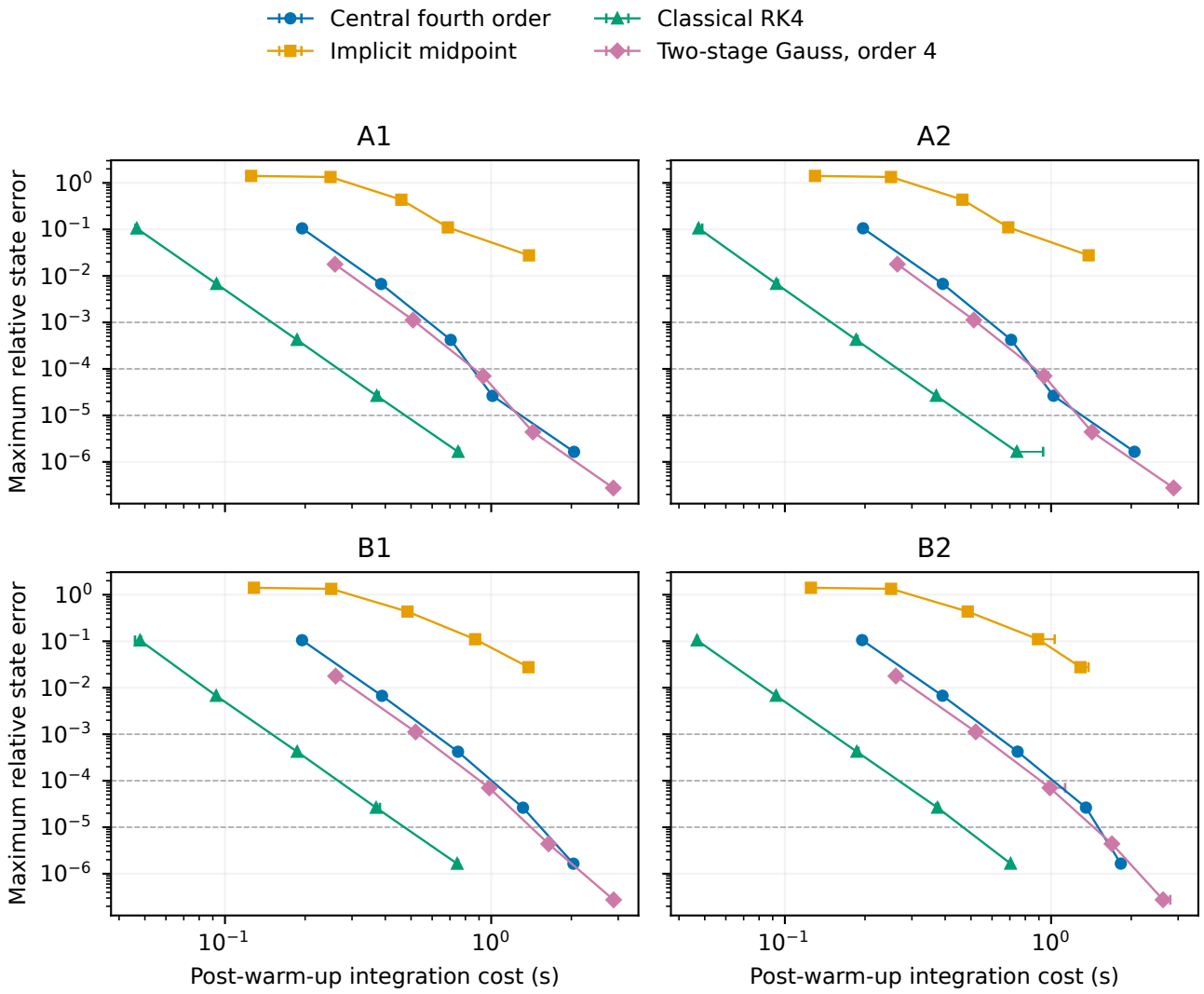



**Figure 1.** State error versus integration cost for all five step sizes and four initial conditions. Method names match the text; $\boldsymbol{A}_1, \boldsymbol{A}_2, \boldsymbol{B}_1, \boldsymbol{B}_2$ are defined in (48). Each point uses the median of three post-warm-up timings; thin horizontal bars show the minimum–maximum range. Dashed lines mark the three prescribed accuracy targets. Shared preparation cost is reported separately and excluded from this axis. All 80 configurations are shown; timing bars are ranges of three runs, not confidence intervals.

**Table 4.** Post-warm-up integration cost ranges in seconds at three state-accuracy targets, over four initial conditions. Selection uses only the five measured step sizes; shared preparation of 57.0300 s is additional.

| Method | $10^{-3}$ | $10^{-4}$ | $10^{-5}$ |
|---|---|---|---|
| Central fourth order | 0.705–0.750 | 1.011–1.352 | 1.828–2.057 |
| Implicit midpoint | Not attained(4) | Not attained(4) | Not attained(4) |
| Classical RK4 | 0.185–0.187 | 0.370–0.374 | 0.704–0.751 |
| Two-stage Gauss, order 4 | 0.933–0.990 | 0.933–0.990 | 1.425–1.692 |

### *4.1 Quartic normal form and complete low-order cancellation*

In the canonical modal coordinates $\boldsymbol{W} = (\boldsymbol{X}, \boldsymbol{Y})$, set

$$\begin{aligned} a_j &= (X_j + \mathrm{i}Y_j)/\sqrt{2}, \quad I_j = a_j \bar{a}_j, \\ \{a_j, \bar{a}_k\} &= -\mathrm{i}\delta_{jk}, \quad H_{2,\mathrm{dec}} = \boldsymbol{\omega} \cdot \boldsymbol{I}. \end{aligned} \tag{50}$$

The real-coordinate Poisson bracket is $\{F, G\} = \nabla F^{\mathsf{T}} \boldsymbol{J} \nabla G$. Here $H_3, H_4$ denote the cubic and quartic homogeneous terms of $H_{\mathrm{dec}} = H_{2,\mathrm{dec}} + H_3 + H_4$; $H_4$ is not the Hessian of a quadratic potential. The modal transformation satisfies (A.2), and the monomial indexing specifies every coefficient. Errors linking this object to the full material model remain separate.

Define $L_\chi F = \{\chi, F\}$. On a monomial $\boldsymbol{a}^{\boldsymbol{\mu}}\bar{\boldsymbol{a}}^{\boldsymbol{\nu}}$, the homological operator acts as

$$\{\boldsymbol{a}^{\boldsymbol{\mu}}\bar{\boldsymbol{a}}^{\boldsymbol{\nu}}, H_{2,\mathrm{dec}}\} = -\mathrm{i}d_{\boldsymbol{\mu\nu}}\boldsymbol{a}^{\boldsymbol{\mu}}\bar{\boldsymbol{a}}^{\boldsymbol{\nu}}, \qquad d_{\boldsymbol{\mu\nu}} = (\boldsymbol{\mu}-\boldsymbol{\nu})\cdot\boldsymbol{\omega}. \tag{51}$$

Treating the supplied decimal frequencies as exact rationals, the minimum cubic absolute divisor is $32.24949540046847$, and the minimum nonzero quartic absolute divisor is $0.27186195033089$. The cubic kernel is empty, while the quartic kernel consists precisely of action monomials with $\boldsymbol{\mu} = \boldsymbol{\nu}$. This finite-order result permits the homological divisions; it does not establish a Diophantine condition for all integer vectors.

Let $\Pi$ project onto the quartic kernel. With zero kernel components in the generators, define

$$\begin{aligned}
(\chi_3)_{\boldsymbol{\mu\nu}} &= -\mathrm{i}(H_3)_{\boldsymbol{\mu\nu}}/d_{\boldsymbol{\mu\nu}},\\
G_4 &= H_4 + \tfrac{1}{2}\{\chi_3, H_3\}, \qquad \widehat{Z}_4 = \Pi G_4,\\
(\chi_4)_{\boldsymbol{\mu\nu}} &= -\mathrm{i}(G_4)_{\boldsymbol{\mu\nu}}/d_{\boldsymbol{\mu\nu}} \quad (d_{\boldsymbol{\mu\nu}} \neq 0),\\
\Pi\chi_4 &= 0.
\end{aligned} \tag{52}$$

The coefficients satisfy the reality conjugacies and define real homogeneous Hamiltonians. The complete transformed polynomial through degree four is

$$\begin{aligned}
[\mathrm{e}^{L_{\chi_4}}\mathrm{e}^{L_{\chi_3}}H_{\mathrm{dec}}]_{\le 4} &= H_{2,\mathrm{dec}} + \big(H_3 + \{\chi_3, H_{2,\mathrm{dec}}\}\big)\\
&\quad + H_4 + \{\chi_3, H_3\}\\
&\quad + \tfrac{1}{2}\{\chi_3, \{\chi_3, H_{2,\mathrm{dec}}\}\}\\
&\quad + \{\chi_4, H_{2,\mathrm{dec}}\}\\
&= H_{2,\mathrm{dec}} + \widehat{Z}_4.
\end{aligned} \tag{53}$$

The cubic parenthesis and the nonkernel quartic terms vanish identically. This statement uses exact finite polynomial algebra with rational decimal inputs, rather than small floating-point residuals or tolerance-based zeroing. If the cubic residual is $E_3 = H_3 + \{\chi_3, H_{2,\mathrm{dec}}\}$, the quartic expression also contains $\frac{1}{2}\{\chi_3, E_3\}$, which cannot be dropped when using rounded coefficients. The stored rounded normal form is not coefficientwise identical to the exact $\widehat{Z}_4$. The proposition uses the latter; Appendix B.4 supplies its coefficients.

Let $\varphi_\chi^t$ be the flow of $\boldsymbol{J}\nabla\chi$. With the stated bracket direction, $\mathrm{e}^{L_\chi}F = F\circ\varphi_\chi^{-1}$, where $-1$ denotes time minus one. Therefore

$$\begin{aligned}
\mathcal{C} &= \varphi_{\chi_3}^{-1}\circ\varphi_{\chi_4}^{-1},\\
H_{\mathrm{dec}}\circ\mathcal{C} &= h_{\mathrm{NF}}(\boldsymbol{I}) + R_{\mathrm{Lie}}(\boldsymbol{W}),\\
h_{\mathrm{NF}}(\boldsymbol{I}) &= \boldsymbol{\omega}\cdot\boldsymbol{I} + \tfrac{1}{2}\boldsymbol{I}^{\mathsf{T}}\widehat{\boldsymbol{T}}\boldsymbol{I}.
\end{aligned} \tag{54}$$

Because $\boldsymbol{X}_{\chi_3} = O(|\boldsymbol{W}|^2)$ and $\boldsymbol{X}_{\chi_4} = O(|\boldsymbol{W}|^3)$, both finite-time flows are analytic, single-valued, invertible and symplectic on a sufficiently small complex ball. Equation (53) then establishes the full analytic remainder $R_{\mathrm{Lie}} = O(|\boldsymbol{W}|^5)$. A quartic input polynomial need not have a finitely truncated transformed remainder. This Lie-transformation remainder does not represent the unknown truncation error of the material model.

## 4.2 Nondegeneracy and qualitative persistence

The exact $\widehat{Z}_4$ defines the constant twist $\widehat{\boldsymbol{T}} = \partial_{\boldsymbol{I}}^2\widehat{Z}_4$. Its exact determinant is negative; a numerical display and an upper bound are

$$\begin{aligned}
\det\widehat{\boldsymbol{T}} &\simeq -1.5902519571118942207\times 10^{61},\\
\left\|\widehat{\boldsymbol{T}}^{-1}\right\|_\infty &\le 3.44184331720408\times 10^{-12}.
\end{aligned} \tag{55}$$

Nondegeneracy follows from exact rational arithmetic and a rational upper bound, not from the displayed precision. The integrable truncation $h_{\mathrm{NF}}$ already has invariant tori at every fixed action with $I_j > 0$:

$$\begin{aligned}
X_j = \sqrt{2I_j}\cos\phi_j, \quad Y_j = -\sqrt{2I_j}\sin\phi_j,\\
\dot{\boldsymbol{\phi}} = \boldsymbol{\omega} + \widehat{\boldsymbol{T}}\boldsymbol{I}.
\end{aligned} \tag{56}$$

These sets remain invariant even at resonant frequencies, although their trajectories need not be dense in four-dimensional tori. KAM theory addresses persistence after adding $R_{\mathrm{Lie}}$, rather than the integrability of the truncation itself.

**Proposition 4.1.** Qualitative local KAM property of the explicit polynomial. *For the exactly defined $H_{\mathrm{dec}}$, a sufficiently small amplitude threshold exists such that scaled copies of a fixed positive-action shell contain a positive-measure family of analytic Lagrangian invariant four-tori arbitrarily close to the origin. Their dynamics have Diophantine frequencies. The proposition specifies neither a numerical threshold, a particular torus, nor a material operating amplitude.*

**Proof.** We use Theorem A in Pöschel's Version 1.3 (2009) (Pöschel, 2001), with the corrected proof of the same theorem by Bounemoura (Bounemoura, 2024). Its four-dimensional analytic parameter formulation uses a common complex domain with action radius $r_*$, angle-strip width $s$, parameter width $\eta$, and $N_{\boldsymbol{\lambda}} = e(\boldsymbol{\lambda}) + \boldsymbol{\lambda}\cdot\boldsymbol{u}$. Taking $\tau = 4 >$

$4-1$, the sufficient conditions include

$$\begin{aligned} |\boldsymbol{k}\cdot\boldsymbol{\lambda}| &\geq \alpha|\boldsymbol{k}|_1^{-4} \quad (\boldsymbol{k}\neq 0),\\ \|P\| &\leq c_*\alpha r_* s^5, \quad \alpha s^5 \leq \eta. \end{aligned} \tag{57}$$

as well as the theorem's domain, real-analyticity and parameter hypotheses. Here $c_* > 0$ is a theorem constant.

Set $\boldsymbol{W} = \varepsilon\boldsymbol{w}$, $\boldsymbol{I} = \varepsilon^2\boldsymbol{A}$, and rescale time by $\sigma = \varepsilon^2 t$. On a fixed positive-action complex shell away from $A_j = 0$, (54) becomes

$$\begin{aligned} K_\varepsilon &= \varepsilon^{-4}(H_{\mathrm{dec}}\circ\mathcal{C})(\varepsilon\boldsymbol{w})\\ &= \varepsilon^{-2}\boldsymbol{\omega}\cdot\boldsymbol{A} + \tfrac{1}{2}\boldsymbol{A}^{\mathsf{T}}\widehat{\boldsymbol{T}}\boldsymbol{A} + P_\varepsilon,\\ &\|P_\varepsilon\| \leq C_*\varepsilon. \end{aligned} \tag{58}$$

This bound uses the entire analytic $O(5)$ remainder, not only degrees five and six. The frequency image is the translated domain $\boldsymbol{b}_\varepsilon + \widehat{\boldsymbol{T}}\boldsymbol{A}$, with $\boldsymbol{b}_\varepsilon = \varepsilon^{-2}\boldsymbol{\omega}$ and a fixed shape determined by invertible $\widehat{\boldsymbol{T}}$. Within a fixed inner domain, the resonant slab $|\boldsymbol{k}\cdot\boldsymbol{\lambda}| < \alpha|\boldsymbol{k}|_1^{-4}$ has volume bounded by a constant times $\alpha|\boldsymbol{k}|_1^{-5}$. The sum over four-dimensional integer vectors converges uniformly in the translation centre. A fixed sufficiently small $\alpha$ therefore leaves a positive-measure Diophantine set in each frequency domain, without assuming that $\boldsymbol{\omega}$ itself satisfies an all-integer condition.

For parameter $\boldsymbol{\lambda}$, set $\boldsymbol{a}_\varepsilon(\boldsymbol{\lambda}) = \widehat{\boldsymbol{T}}^{-1}(\boldsymbol{\lambda} - \boldsymbol{b}_\varepsilon)$ and $\boldsymbol{A} = \boldsymbol{a}_\varepsilon + \boldsymbol{u}$. Relative to the linear $N_{\boldsymbol{\lambda}}$, the perturbation must include

$$\begin{aligned} P &= \tfrac{1}{2}\boldsymbol{u}^{\mathsf{T}}\widehat{\boldsymbol{T}}\boldsymbol{u} + P_\varepsilon(\boldsymbol{a}_\varepsilon + \boldsymbol{u}, \boldsymbol{\phi}),\\ \|P\| &\leq Mr_*^2 + C_*\varepsilon, \quad M = \tfrac{1}{2}\sum_{ij}|\widehat{T}_{ij}|. \end{aligned} \tag{59}$$

First choose common $s, \eta, \alpha$ and sufficiently small $r_*$ so that parameter centres remain in the domain, $\alpha s^5 \leq \eta$, and $Mr_*^2 \leq c_*\alpha r_* s^5/2$. Then decrease $\varepsilon$ to satisfy the remaining budget. Theorem A and its parameter control yield a family of tori. The nondegenerate frequency map converts parameters back into actions of the original Hamiltonian while retaining positive measure. The analytic canonical chart $\mathcal{C}$ and the nonzero scaling transport the tori to the original coordinates; physical-time frequencies are $\varepsilon^2\boldsymbol{\lambda}$. This establishes qualitative existence, not a computed amplitude certificate. □

This is an application of a classical nondegenerate KAM theorem to an explicit polynomial, not a new KAM theorem. The fixed quadratic $H_2$ has zero twist and cannot enter the same nondegenerate argument directly; this does not mean that its linear flow lacks invariant tori.

### 4.3 Links to learning, discretization and material applications

A sampled exact-flow torus remains invariant as a set, whereas an algorithmic map requires its own proof. For example, the specified-map theorem of Figueras–Haro–Luque (Theorem 2.6, author manuscript of 26 August 2016) requires an analytic exact symplectic map, step-frequency nonresonance and controlled embedding/torsion/residual bounds (Figueras et al., 2017). These data are not supplied by the present integration or prediction errors. The retained conclusion is the qualitative family for the exact $H_{\mathrm{dec}}$ flow; the table specifies where transfer would require new evidence.

## 5 Guidance for modelling and numerical choices

### 5.1 Choose the prior and representation in separate comparisons

A practical first step is to include the fixed $H_2$ in a controlled comparison under the intended data and state conventions. The stable prior directions in Section 3.3 support this choice for the present task. Retaining the no-prior alternatives makes it possible to check whether the prescribed energy remains useful when conditions change.

Next, compare representations within each prior level. The real gray-box model is a useful initial comparator here: the sealed analysis left the gray-box contrast unresolved, while the new group/seed sensitivity analysis favours GR. The unresolved no-prior contrast and interaction do not establish equivalence or a change in representation effect caused by the prior. Width, weight constraints and optimization remain part of the comparison, so selection should use measured prediction and resource costs rather than the real/complex label alone. The five-model descriptive table supplies context; it does not replace the paired effects.

Before deployment, translate the normalized state metric in (46) into the relevant engineering observables and acceptable errors. It is not directly a tip-position, contact-force or control-success metric. Fixed-$H_2$ error and learned-energy drift also measure different quantities and should remain separate.

### 5.2 Select the integrator against explicit accuracy and energy requirements

Prescribe a state-error target, retain all measured configurations, and compare energy changes and full cost among those that attain it. On the tested polynomial, RK4 costs

**Table 5.** Object-specific scope of the KAM conclusion. Uncertified links remain conditional.

| Object | Established conclusion and boundary |
|---|---|
| Exact $H_{\mathrm{dec}}$ flow | The qualitative family in Proposition 4.1; no numerical certificate for a specified amplitude or torus. |
| Learned $H_{\boldsymbol{\theta}}$ | Requires a common true Hamiltonian, canonical chart, energy gauge and analytic error. Map loss, statistical intervals and network AD do not supply these quantities. |
| Material reference | Requires sources for the original fit, mass and equilibrium, low-order input errors and the full material truncation remainder. $R_{\mathrm{Lie}}$ cannot replace them. |
| Specified map | The exact central fourth-order root is a candidate exact symplectic map; a common complex domain, step nonresonance, averaged torsion and residual bounds remain required. Actual RK4 is not generally exact symplectic. |

less than central fourth order at the selected state targets, whereas the central method has smaller energy changes. Against Gauss, the ordering changes with the target and the central method's faster $10^{-3}$ configurations have larger energy changes. Table 4 and Section 3.4 retain all three targets, midpoint's nonattainment and the additional 57.0300 s preparation cost. A numerical choice therefore depends on the stated energy requirement as well as state accuracy and budget.

These comparisons concern $H_{\mathrm{dec}}$. Changing the integrator used with a trained $H_{\boldsymbol{\theta}}$ would require a comparison on that learned field, including its derivative and implicit-solve costs; the current learning results use RK4.

### 5.3 Use conservative structure at the scale supported by the model

For the same explicit polynomial, Proposition 4.1 explains a qualitative family of quasiperiodic motions near the equilibrium. This is useful structural interpretation, not a numerical operating amplitude. Table 5 specifies why transfer to the material or learned system needs additional evidence. In particular, the full material error is not the Lie remainder of the defined polynomial.

Modal order and forcing also require application-specific checks. The auxiliary representation record, retained with the technical material, compares three trajectories against a finite $N = 6$ reference; it does not establish predictive sufficiency of $N = 2$, an optimal order or a safety region. The supported workflow is thus to test the prior, compare representations, and select time stepping under explicit accuracy, energy and cost criteria before drawing device-level conclusions.

## 6 Conclusions

The fixed reduced-catheter design separates prior information from real/complex representation. Both prior effects retain their improvement direction across the prescribed sensitivity procedures. The sealed analysis did not resolve additional complex benefit; the post-experiment group/seed analysis gives a negative gray-box real-minus-complex contrast, favouring the real implementation. No-prior representation and interaction remain unresolved. These are finite-design findings under the stated sampling assumptions, not universal superiority, equivalence or certified population coverage.

On the independent weakly nonlinear $H_{\mathrm{dec}}$, the existing central fourth-order method trades smaller observed energy changes for approximately 2.59–4.02 times RK4's integration cost at selected state targets. It is faster than Gauss only at $10^{-3}$ on the fixed grid, with larger energy changes; Gauss is faster at $10^{-4}$ and $10^{-5}$. Preparation is additional, and midpoint's nonattainment is limited to this grid. The comparison establishes no consistent performance advantage.

For that same polynomial, complete quartic cancellation, the analytic Lie remainder and nonsingular twist support a qualitative local KAM family at sufficiently small amplitudes. This is an application of classical theory, separate from the numerical trade-off and from the learning results. It provides no learned/material or specified-amplitude certificate. The core increment is the task-specific separation and sensitivity of prior/representation effects; the method derivations and conservative-model results support a workflow that tests the prior, evaluates representation separately, and chooses integration jointly by state accuracy, energy behaviour and full cost.

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### *Data availability*

Coefficient files, complete comparison tables, source indexes and build files accompany the review package. Repository location, access terms, ownership and permission to share remain to be confirmed by the authors.

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# A Supplementary derivations for central schemes and complex generation

## A.1 Coefficients and indices of the specified polynomial

The complete coefficients in (5) accompany the supplementary material as `hdec-coefficients.json`. The fields `omega`, `H3`, and `H4` contain the quadratic modal frequencies and cubic and quartic monomial coefficients. The frequencies are

$$\boldsymbol{\omega} = \begin{pmatrix} 34.95334005159986 \\ 35.22520200193075 \\ 217.78628967842488 \\ 220.49013432955627 \end{pmatrix} \mathrm{s}^{-1}. \tag{A.1}$$

The JSON uses sorted, zero-based variable indices. For example, the key `(0,0,1)` denotes the coefficient of $W_1^2 W_2$, not an exponent vector. Each sorted monomial is counted once, with no additional permutation factor or factorial division. Explicit zero entries are supplied coefficients, not replacements for missing physical input. The cubic and quartic coefficients come from the stored modal polynomial; the quadratic term is defined by (5), and the constant and linear terms of this specified polynomial are zero. This definition does not eliminate material equilibrium errors or identify a quadratic-potential Hessian as a quartic term.

The source modal chart uses $\delta\boldsymbol{q} = \boldsymbol{L}_q \boldsymbol{X}$ and $\boldsymbol{p} = \boldsymbol{L}_p \boldsymbol{Y}$ with $\boldsymbol{L}_p = \boldsymbol{L}_q^{-\mathsf{T}}$, so

$$\operatorname{diag}(\boldsymbol{L}_q, \boldsymbol{L}_p)^{\mathsf{T}} \boldsymbol{J} \operatorname{diag}(\boldsymbol{L}_q, \boldsymbol{L}_p) = \boldsymbol{J}. \tag{A.2}$$

The variable $\boldsymbol{Y}$ is the paired momentum in this separate chart. Errors connecting numerical fitting, recentering, and linear diagonalization to the complete material model differ from the normal-form remainder of the defined polynomial. Storing more coefficient digits does not remove the limitations on material identification.

## A.2 Generating coefficients and higher-derivative assembly

The explicit multilinear version of (15) is

$$S_{k+1} = \frac{1}{k+1} \sum_{r=1}^{k} \frac{1}{2^r r!} \sum_{\substack{j_1+\cdots+j_r=k \\ j_i \geq 1}} H^{(r)}[\boldsymbol{J}\nabla S_{j_1}, \ldots, \boldsymbol{J}\nabla S_{j_r}]. \tag{A.3}$$

Only already determined coefficients appear on the right. For the fourth-order update, put $\boldsymbol{B} = H''$, $\boldsymbol{T}_3 = H'''$, $\boldsymbol{T}_4 = H''''$, and $\boldsymbol{f} = \boldsymbol{J}\nabla H$. For directions $\boldsymbol{a}, \boldsymbol{b}$,

$$\mathrm{D}\boldsymbol{f}[\boldsymbol{a}] = \boldsymbol{J}\boldsymbol{B}\boldsymbol{a}, \qquad D^2\boldsymbol{f}[\boldsymbol{a},\boldsymbol{b}] = \boldsymbol{J}\boldsymbol{T}_3[\,\cdot\,,\boldsymbol{a},\boldsymbol{b}], \tag{A.4}$$

$$DE[\boldsymbol{a}] = \boldsymbol{T}_3[\boldsymbol{a},\boldsymbol{f},\boldsymbol{f}] + 2\boldsymbol{B}[\mathrm{D}\boldsymbol{f}[\boldsymbol{a}],\boldsymbol{f}], \tag{A.5}$$

$$\begin{aligned} D^2E[\boldsymbol{a},\boldsymbol{b}] = {} & \boldsymbol{T}_4[\boldsymbol{a},\boldsymbol{b},\boldsymbol{f},\boldsymbol{f}] + 2\boldsymbol{T}_3[\boldsymbol{a},\mathrm{D}\boldsymbol{f}[\boldsymbol{b}],\boldsymbol{f}] + 2\boldsymbol{T}_3[\boldsymbol{b},\mathrm{D}\boldsymbol{f}[\boldsymbol{a}],\boldsymbol{f}] \\ & + 2\boldsymbol{B}[D^2\boldsymbol{f}[\boldsymbol{a},\boldsymbol{b}],\boldsymbol{f}] + 2\boldsymbol{B}[\mathrm{D}\boldsymbol{f}[\boldsymbol{a}],\mathrm{D}\boldsymbol{f}[\boldsymbol{b}]]. \end{aligned} \tag{A.6}$$

Thus $\nabla S_h^{[3]} = h\nabla H + h^3\nabla E/24$ and $\nabla^2 S_h^{[3]} = h\boldsymbol{B} + h^3\nabla^2 E/24$, specifying the complete Newton matrix. Even when $H$ is quartic, $E$ generally is not, and its terms cannot be truncated according to the degree of the original potential. Network derivatives require a retained higher-order graph starting from real state leaves; detached trajectory outputs do not provide higher derivatives along the original graph.

## A.3 Residual-value control does not ensure structure

For the $C^1$ execution map and state residual in (43), use the common-domain and invertible-branch assumptions of Section 3.1. Differentiating that residual gives the following complete defect identity.

Set $\boldsymbol{E}_r = \mathrm{D}\boldsymbol{r}_h$, evaluate $\boldsymbol{B}, \boldsymbol{C}$ at the computed centre, and put $\boldsymbol{R}_0 = \boldsymbol{B}^{-1}\boldsymbol{C}$. Then

$$\begin{aligned} \mathrm{D}\widetilde{\Psi}_h &= \boldsymbol{R}_0 + \boldsymbol{B}^{-1}\boldsymbol{E}_r, \\ (\mathrm{D}\widetilde{\Psi}_h)^{\mathsf{T}}\boldsymbol{J}\mathrm{D}\widetilde{\Psi}_h - \boldsymbol{J} &= \boldsymbol{R}_0^{\mathsf{T}}\boldsymbol{J}\boldsymbol{B}^{-1}\boldsymbol{E}_r + (\boldsymbol{B}^{-1}\boldsymbol{E}_r)^{\mathsf{T}}\boldsymbol{J}\boldsymbol{R}_0 \\ &\quad + (\boldsymbol{B}^{-1}\boldsymbol{E}_r)^{\mathsf{T}}\boldsymbol{J}(\boldsymbol{B}^{-1}\boldsymbol{E}_r). \end{aligned} \tag{A.7}$$

A residual-value tolerance alone is insufficient. For $H = 0$, the map $(q, p) \mapsto (q + \varepsilon \sin(q/\varepsilon^2), p)$ has residual amplitude at most $\varepsilon$, but area defect $\cos(q/\varepsilon^2)/\varepsilon$. Iteration-count changes and floating-point rounding can also violate the assumed $C^1$ regularity.

### A.4 Fixed-coordinate counterexamples and alternative generation rules

Proposition 2.1 tests $\boldsymbol{r} \cdot \mathrm{d}\boldsymbol{P}$, not the work one-form $\boldsymbol{r} \cdot \mathrm{d}\boldsymbol{Q}$. For $r = -kQ$, choosing $G = -kQP$ gives $G_P = -kQ$ but also $G_Q = -kP$, incorrectly changing the position equation. The force can instead be included once through the real potential $kQ^2/2$; this is not an imaginary-part representation with the original $H$ fixed.

Position-dependent damping $r = -(1 + Q^2)P$ fails similarly: $G = -(1 + Q^2)P^2/2$ has nonzero $G_Q = -QP^2$. In several degrees of freedom, passive damping need not be integrable. For example,

$$\boldsymbol{M} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad \boldsymbol{C} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad \boldsymbol{C}\boldsymbol{M}^{-1} = \begin{pmatrix} 2 & 1/2 \\ 1 & 1 \end{pmatrix}. \tag{A.8}$$

The port $\boldsymbol{r} = -\boldsymbol{C}\boldsymbol{M}^{-1}\boldsymbol{P}$ has power $-(\boldsymbol{M}^{-1}\boldsymbol{P})^{\mathsf{T}}\boldsymbol{C}(\boldsymbol{M}^{-1}\boldsymbol{P}) \le 0$, but $\mathrm{D}_{\boldsymbol{P}}\boldsymbol{r}$ is nonsymmetric, precluding the required $G$ in fixed coordinates. Mass normalization makes the new damping $\boldsymbol{M}^{-1/2}\boldsymbol{C}\boldsymbol{M}^{-1/2}$ symmetric by changing the metric associated with the complex structure; it does not invalidate this fixed-coordinate counterexample.

The zero-power gyroscopic port $\boldsymbol{r} = (-P_2, P_1)$ also has a nonsymmetric Jacobian. On the punctured momentum plane, $\boldsymbol{r} = (-P_2, P_1)/(P_1^2 + P_2^2)$ is locally closed but integrates to $2\pi$ around the unit circle, excluding a single-valued global potential. Power, local integrability, and global single-valuedness are distinct conditions.

For $M(q)$, pointwise substitution of its square root does not guarantee a canonical transformation. Attempting to integrate $\mathrm{D}\chi = \mathrm{diag}(1, 1 + q_1)$, for example, gives incompatible mixed derivatives from $\partial_2\chi_2 = 1 + q_1$ and $\partial_1\chi_2 = 0$. In one degree of freedom, $Q = \int \sqrt{M(q)}\mathrm{d}q$ and $P = p/\sqrt{M(q)}$ are canonical, but the new damping rate $C(q)/M(q)$ may still depend on position and violate the proposition. This only rules out the stated construction; other local coordinate choices would require their own proof.

## B Supplementary numerical analysis, comparison protocols and KAM conditions

Complete tables and their inputs accompany this supplement in `data-analysis`. Its `SOURCE_INDEX.json` lists relative file locations and provenance, including the sealed intervals, all sensitivity outputs and all integration configurations. The polynomial and exact normal-form JSON files accompany Appendix A.1; the package README gives the independent manuscript-build entry.

### B.1 Network families and high-order error norms

The growing-family statement in Proposition 3.3 uses the smooth approximation hypotheses of Hornik (1991), Theorem 3 (Hornik, 1991): fixed finite $m$, sufficiently many continuous activation derivatives, bounded nonconstant $\tanh$, and a target with a $C^m$ extension to a neighbourhood of the compact set. First approximate the function and its derivatives through order $m$ with one hidden layer. Insert $v \mapsto \tanh(\epsilon v)/\epsilon$ on the bounded intermediate range; this tends to the identity in $C^m$. Adjacent weights absorb the scales of the two additional hidden layers, and the real-part embedding preserves this finite-order density. Width and weights may depend on accuracy. This proves no rate for the fixed width 45 model, training convergence, or parameter-budget advantage.

For eight real state variables, put $e = H_{\boldsymbol{\theta}} - H_*$. A common-domain bound $\|e\|_{C^{r+1}} \le \epsilon_{r+1}$ implies

$$\|\boldsymbol{J}\nabla H_{\boldsymbol{\theta}} - \boldsymbol{J}\nabla H_*\|_{C^r} \le c_r \epsilon_{r+1}. \tag{B.1}$$

The norm constant depends on dimension and convention. The central update uses third derivatives of $H$ and its Newton matrix uses fourth derivatives; a retained real-input AD graph with `create_graph` supplies these derivatives. It does not supply their error relative to a true Hamiltonian. No Sobolev or analytic-strip error certificate is assumed in the results of this paper.

## B.2 Statistical estimands, resampling and complete result files

Let $X_{msi}$ be the stored primary metric for model $m$, seed $s$ and trajectory $i$, $Y_{mi} = \mathrm{median}_s\, X_{msi}$, and $\mathrm{D}_i = Y_{Ai} - Y_{Bi}$. The two finite-design estimands are

$$\mu_T = \frac{1}{16}\sum_i D_i, \qquad \mu_G = \frac{1}{6}\sum_{g=1}^{6}\frac{1}{n_g}\sum_{i\in g}\mathrm{D}_i. \tag{B.2}$$

Replacing seed medians with means changes the estimand rather than correcting the sealed table. Leave-one checks remove each of six whole groups, five common seed labels, and 20 individual model–seed combinations. Remaining group/trajectory weights are renormalized, and the median of an even number of seeds averages the middle two. Common-seed deletion is an influence analysis, not evidence of common-random-number training across models.

Each of three schemes uses 20,000 replicates: groups only, seeds only, and independently combined groups and seeds. Whole-group draws are shared across models and retain every within-group phase. Five seed labels are sampled with replacement independently for each model, and that draw is used across all its trajectories. The joint scheme draws groups first, then seeds in R,GR,C,GC order; the seed-only scheme consumes no group draws. Separate PCG64 streams start at 2026090901, 2026090902, 2026090903, respectively, after lexicographic sorting of group, seed and trajectory labels. All effects and both weights share each model draw, and the interaction is computed from the same four-model comparisons.

Intervals use linear percentiles 2.5%/97.5%; the nominal five-effect adjustment uses 0.5%/99.5%. The latter depends on individual coverage and the working sampling model and is not a simultaneous-coverage certificate. For trajectory weighting, the joint-bootstrap denominator is the actual number of trajectories in the sampled whole groups and varies by replicate. Group weighting averages the six sampled group means. The 20 point estimates for four aggregation targets, 620 leave-one rows and 30 interval rows are retained in the supplementary data index. These new outputs do not overwrite the sealed intervals. The main table uses the original E1 simple effects and corresponding interaction, without mixing in the E2 gray-box interval generated by a different random stream.

All deterministic prior-effect directions are positive and the primary adjusted intervals have positive lower endpoints. All six new gray-box representation intervals are negative; all six no-prior and interaction intervals cross zero. These statements concern the prescribed finite-design sensitivity analysis, not new training sets, new orbits or population risk. The $H_2$ energy error measures a fixed quadratic comparator, whereas learned-$H_{\boldsymbol{\theta}}$ drift concerns another scalar. A constant shift leaves dynamics unchanged but can change a relative metric divided by $|H_{\boldsymbol{\theta}}(0)|$; float32/float64 evaluation may additionally suffer cancellation. This resolution boundary is retained without substituting the de-constantized diagnostic for the formal metric.

## B.3 Integrator implementation, cost and complete energy summary

Let $C_3, C_4$ be sums of absolute monomial coefficients of the specified homogeneous polynomials. The protocol fixes amplitudes by $C_3R + C_4R^2 = 0.05\omega_{\min}$ and $r = R\sqrt{\omega_{\min}/(8\omega_{\max})}$. This restricts a local energy range for the exact conservative flow, not discrete trajectories or a certified KAM amplitude. All 80 configurations, cost selections and failure fields are retained as separate supplementary CSV files. Figures and tables use formal method names.

The two-stage Gauss stage states $\boldsymbol{Z}_i \in \mathbb{R}^8$ satisfy

$$\begin{aligned} \boldsymbol{F}_i(\boldsymbol{Z}) &= \boldsymbol{Z}_i - \boldsymbol{W}_n - h\sum_{j=1}^{2} a_{ij}\boldsymbol{f}(\boldsymbol{Z}_j) = 0, \\ \mathrm{D}_{\boldsymbol{Z}_j}\boldsymbol{F}_i &= \delta_{ij}\boldsymbol{I} - ha_{ij}\boldsymbol{J}H''(\boldsymbol{Z}_j), \qquad \boldsymbol{W}_{n+1} = \boldsymbol{W}_n + \tfrac{h}{2}(\boldsymbol{f}(\boldsymbol{Z}_1) + \boldsymbol{f}(\boldsymbol{Z}_2)), \\ (a_{ij}) &= \begin{pmatrix} 1/4 & 1/4 - \sqrt{3}/6 \\ 1/4 + \sqrt{3}/6 & 1/4 \end{pmatrix}. \end{aligned} \tag{B.3}$$

All 16 unknowns and residual components have state units $\boldsymbol{W}$; central fourth order and midpoint use eight endpoint-state unknowns. All three implicit methods require infinity norms of the residual and computed final Newton correction, divided by $R$, at most $10^{-13}$. At most 20 corrections are permitted, with trial factors $1, 1/2, \ldots, 1/128$. Tightening to $10^{-14}$ at the finest step for every initial condition and implicit method gives 12 checks, all satisfying the requirement that changes be below 10% of the relevant main error/threshold. This does not bound residual derivatives.

Baseline DOP853 uses relative tolerance $10^{-11}$, absolute tolerance $R10^{-13}$ and maximum step $h_4/4$. The fine reference uses $2 \times 10^{-13}$, $2R10^{-15}$ and $h_4/8$. Comparisons on common grids include dense output. Differences between the two references for each of four initial conditions range from $2.31 \times 10^{-14}$ to $4.26 \times 10^{-14}$, below 1% of the accuracy targets and 10% of the relevant method errors; the prescribed third level was not triggered. The largest state change in the 12 solver-tightening checks is approximately $1.02 \times 10^{-9}$. These are sensitivity diagnostics, not rigorous exact-flow enclosures, and do not remove float64 input rounding.

Each configuration is warmed once and timed three times in a fixed randomized order. Limits are six hours overall, 600 s per configuration and 8 GiB memory; no execution or resource failures occurred. Recorded work includes $H$ values, gradients, Hessians, third- and fourth-order contractions, linear solves, Newton corrections, line searches and residual diagnostics. These operations and native-step energy evaluations are included in timing, rather than comparing only steps or vector-field calls. Table B.1 retains all methods and five step sizes. Each row covers four initial conditions: all completed 4/4 and none failed reference/solver resolution. Missing an accuracy target remains distinct from execution failure.

**Table B.1.** Energy-change ranges over four initial conditions at every step size. All rows completed 4/4 with no unresolved reference or solver configurations.

| Method | $j$ | $\min e_H$ | $\max e_H$ |
|---|---|---|---|
| Central fourth order | 0 | $2.206\,10^{-8}$ | $6.547\,10^{-8}$ |
| Central fourth order | 1 | $1.429\,10^{-9}$ | $4.206\,10^{-9}$ |
| Central fourth order | 2 | $1.942\,10^{-10}$ | $2.781\,10^{-10}$ |
| Central fourth order | 3 | $2.215\,10^{-11}$ | $7.274\,10^{-11}$ |
| Central fourth order | 4 | $2.242\,10^{-12}$ | $1.022\,10^{-11}$ |
| Implicit midpoint | 0 | $5.775\,10^{-7}$ | $1.733\,10^{-6}$ |
| Implicit midpoint | 1 | $1.489\,10^{-7}$ | $4.445\,10^{-7}$ |
| Implicit midpoint | 2 | $3.748\,10^{-8}$ | $1.119\,10^{-7}$ |
| Implicit midpoint | 3 | $9.405\,10^{-9}$ | $2.801\,10^{-8}$ |
| Implicit midpoint | 4 | $2.351\,10^{-9}$ | $7.009\,10^{-9}$ |
| Classical RK4 | 0 | $7.991\,10^{-2}$ | $8.030\,10^{-2}$ |
| Classical RK4 | 1 | $2.655\,10^{-3}$ | $2.669\,10^{-3}$ |
| Classical RK4 | 2 | $8.338\,10^{-5}$ | $8.382\,10^{-5}$ |
| Classical RK4 | 3 | $2.608\,10^{-6}$ | $2.622\,10^{-6}$ |
| Classical RK4 | 4 | $8.152\,10^{-8}$ | $8.195\,10^{-8}$ |
| Two-stage Gauss, order 4 | 0 | $4.068\,10^{-9}$ | $1.099\,10^{-8}$ |
| Two-stage Gauss, order 4 | 1 | $2.565\,10^{-10}$ | $6.849\,10^{-10}$ |
| Two-stage Gauss, order 4 | 2 | $1.554\,10^{-11}$ | $4.457\,10^{-11}$ |
| Two-stage Gauss, order 4 | 3 | $5.133\,10^{-11}$ | $8.429\,10^{-11}$ |

Continued on next page

Table B.1 (continued)

| Method | $j$ | $\min e_H$ | $\max e_H$ |
|---|---|---|---|
| Two-stage Gauss, order 4 | 4 | $1.971\,10^{-12}$ | $1.059\,10^{-11}$ |

## B.4 Exact normal-form coefficients and source boundaries

Table B.2 displays the ten coefficients in $\widehat{Z}_4 = \sum_{|\beta|=2} c_\beta \boldsymbol{I}^\beta$. Full rationals in `normal-form-exact.json` are extracted directly from the existing exact-cancellation result, without repeating the normal-form computation. That algebra used $\zeta_j = X_j + \mathrm{i}Y_j = \sqrt{2}a_j$, $\{\zeta_j, \bar{\zeta}_k\} = -2\mathrm{i}\delta_{jk}$ and $I_j = \zeta_j\bar{\zeta}_j/2$, equivalent to the convention in the main text. Bracket coefficients from the two conventions must not be mixed. Display decimals do not replace rational coefficients.

**Table B.2.** Display coefficients of the exact quartic normal form. Exponents are ordered as $I_1, I_2, I_3, I_4$; exact rationals are supplied separately.

| $\beta$ | $c_\beta$ |
|---|---|
| $(2,0,0,0)$ | $434470615578.06039327$ |
| $(1,1,0,0)$ | $583279003999.93543432$ |
| $(1,0,1,0)$ | $3734689434505.0453818$ |
| $(1,0,0,1)$ | $-4585094327803.5669993$ |
| $(0,2,0,0)$ | $440827566852.01874934$ |
| $(0,1,1,0)$ | $-2375967766950.0738049$ |
| $(0,1,0,1)$ | $12143155842211.582250$ |
| $(0,0,2,0)$ | $53060627601243800.310$ |
| $(0,0,1,1)$ | $-1633954193532226930.8$ |
| $(0,0,0,2)$ | $-1.634506022020102537\,10^{20}$ |

The modal cubic and quartic inputs come from the recorded common-canonical-chart coefficient source; $H_{\mathrm{dec}}$ is defined by those inputs. The exact determinant sign and inverse bound apply to $\widehat{\boldsymbol{T}}$ obtained by cancellation under that definition. The relative coefficient difference from the stored rounded normal form is approximately $3.29 \times 10^{-17}$, still not exact equality. Original material-fit, mass and centring errors do not vanish as a result. Material error can include degrees zero through four and a higher-order tail; the Lie remainder starts at degree five for the explicit polynomial and exact chart. These require different evidence.